20

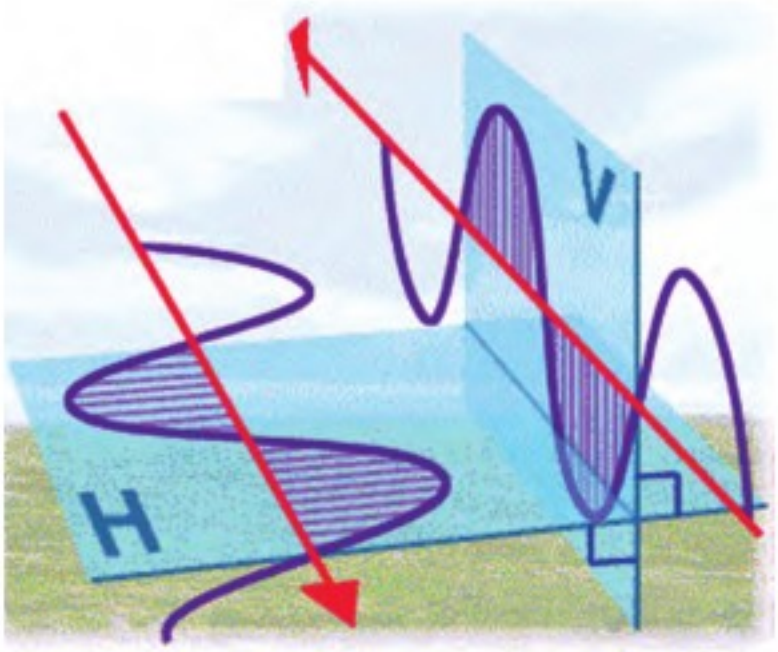
CHAPTER



Polarization

20.1. INTRODUCTION

Interference and diffraction phenomena proved that light is a wave motion and enabled the determination of the wavelength. However, they do not give any indication regarding the character of the waves. Whether the light waves are longitudinal or transverse, or whether the vibrations are linear or circular cannot be deduced from the above two phenomena, as all kinds of waves under suitable conditions exhibit interference and diffraction. In 1816 Arago and Fresnel showed that light waves vibrating in mutually perpendicular planes do not interfere. In 1817 Thomas Young postulated that light waves are *transverse waves* and explained



Polarization

At a Glance

- Introduction
- Unpolarized and Polarized Light
- Types of Polarization
- Polaroid Sheets
- Malus' Law
- Double Refraction in Calcite Crystal
- Effect of Polarizer on Light of Different Polarizations
- Experimental Determination of Principal Refractive Indices
- Phase Difference Between E-ray and O-ray
- Retarders
- Production of Circularly Polarized Light
- Babinet Compensator
- Double Image Polarizing Prisms
- Optical Activity
- Specific Rotation
- Experimental Verification of Fresnel's Theory
- Biquartz
- Electro-optic and Magneto-optic Effects
- Magneto-optic Effects
- Anisotropy Induced by Mechanical Strain
- Photoelasticity

the absence of interference between light waves polarized in mutually perpendicular planes. Thus, the existence of polarization property is a direct consequence of light being a transverse wave. Light coming from common light sources is unpolarized. It can be transformed into different types of polarization using optical devices. The state of polarization cannot be perceived by an unaided human eye. An understanding of polarization is essential for understanding the propagation of electromagnetic waves guided through wave-guides and optical fibres. Polarized light has many important applications in industry and engineering. One of the most important applications is in liquid crystal displays (LCDs), which are widely used in wristwatches, calculators, TV screens, etc.

20.2. POLARIZATION

Waves are basically of two types: (*i*) longitudinal waves and (*ii*) transverse waves.

(*i*) A wave in which particles of the medium oscillate to and fro along the direction of propagation is called a **longitudinal wave**. Waves produced on a spring and sound waves are examples of longitudinal waves. The longitudinal wave consists of alternate compressions and rarefactions, as shown in Fig.20.1 (c) & (d).

(*ii*) A wave in which every particle of the medium oscillates up and down at right angles to the direction of wave propagation is called a **transverse wave**. Ripples on water surface and waves on a rope are examples of transverse waves. The wave propagates in the form of alternating crests and troughs, as shown in Fig.20.1 (a) & (b).

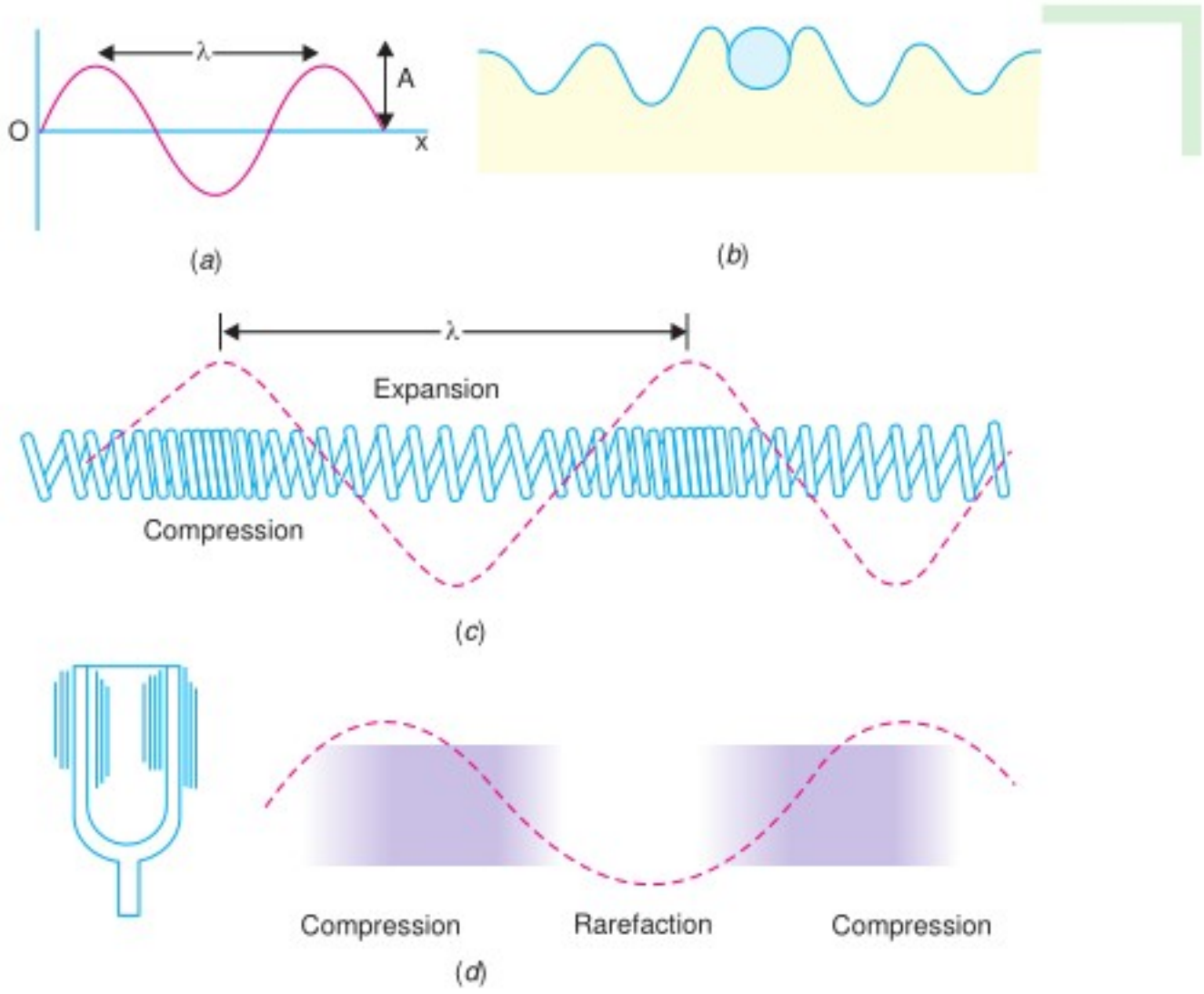


Fig. 20.1

In a longitudinal wave, all directions perpendicular to the wave propagation are equivalent. On the other hand, a preferential direction normal to the wave propagation exists in a transverse wave.

The preferential direction in a transverse wave is the direction of vibration of the particles and it differs from all other directions. The existence of a preferential direction for a transverse wave leads to the characteristic phenomenon known as polarization. Polarization is not found with longitudinal waves as they do not possess a directional property. Thus, polarization is specific to transverse waves.

Light waves are transverse waves consisting of electric and magnetic fields vibrating perpendicular to each other and to the direction of propagation. The vibrating electric field vector and the direction of propagation of the wave constitute a plane. There is an infinite number of such planes around the direction of propagation. In an ideal light wave, the vibrations of electric vector are confined to a single plane. In practice, light sources emit a mixture of light waves whose planes of vibration are randomly oriented about the direction of propagation. Such random orientation of vibration planes gives rise to symmetry about the wave propagation direction. As a result, the transverse nature of the wave gets concealed. The process of removing the symmetry and bringing in one-sidedness in the light wave is called **polarization**.

20.3. UNPOLARIZED AND POLARIZED LIGHT

Light wave is a transverse electromagnetic wave made up of mutually perpendicular, fluctuating electric and magnetic fields. Fig.20.2 (*a*) shows the electric field in the *xy*- plane, the magnetic field in the *xz*-plane and the propagation of the wave in the *x*- direction. The right-hand part of the diagram shows the variation of the electric field in space as the wave propagates. Traditionally, light wave is described by the **electric field vector**, ***E***, and accordingly, only the electric field vector is shown in Fig.20.2 (*b*).

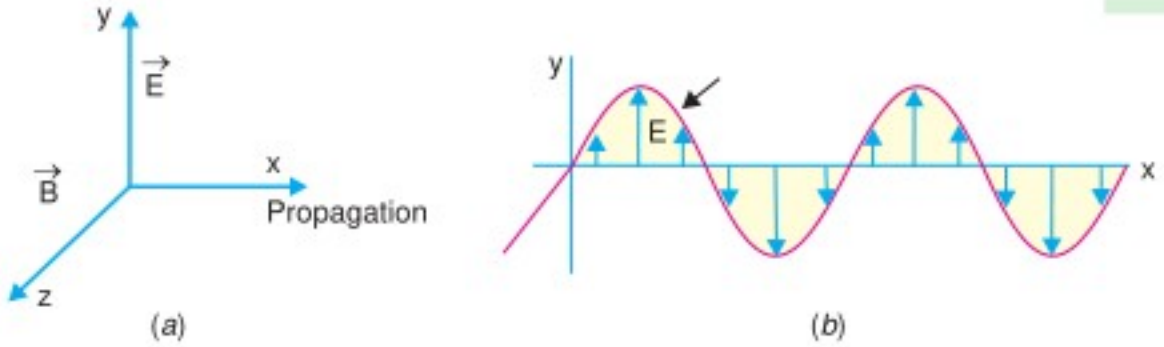


Fig. 20.2: A light wave is described by the electric field vector, *E*

As the electric field is a vector, it points in a particular direction in space. The **polarization** of an electromagnetic wave refers to the orientation of its electric field vector *E*. If we could view a light wave coming from an ordinary source towards us, then we would observe that the direction of *E* is randomly varying with time on a very fast scale. The light from an incandescent bulb, for example, emits a mixture of light waves with electric field components that change randomly on a scale of 10^{-14} s, almost as fast as the optical frequency itself. As a result, the **direction of oscillation of the electric field vector in an ordinary light beam occurs in all the possible planes perpendicular to the beam direction,** as illustrated in Fig.20.3. A light wave, in which **E**-vector oscillates in more than one plane, is referred to as **unpolarized light**. Light emitted by the sun, by an incandescent lamp, or by a candle flame is unpolarized light.

Polarized Light

Polarized light is not produced naturally. It is obtained by converting natural light into polarized light using optical elements. The process of transforming unpolarized light into polarized light is polarization. A polarized light wave is a light wave with a definite direction of oscillation of the *E*- vector, which occurs in a *single plane* or in *some specific way*. For example, the wave in Fig.20.2 (b) is a polarized wave. **Polarized light** is the light that contains waves that only fluctuate in one specific plane.

We designate the plane created by the direction of oscillation of the electric field vector $\boldsymbol{E}$ and the direction of propagation of the beam as the **plane of polarization of light wave**. Thus, the *xy*-plane is the plane of polarization in Fig.20.2 (*b*).

20.4. NATURAL LIGHT IS UNPOLARIZED LIGHT

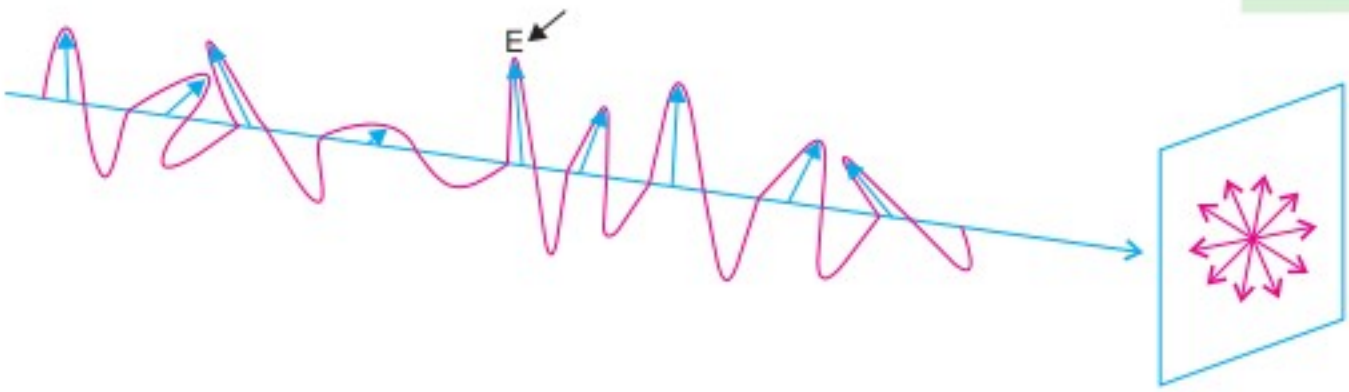


Fig. 20.3: Natural light is unpolarized

We mentioned earlier that light from the ordinary light sources is unpolarized. Let us find out the reason. We know that atoms emit light. Any light source consists of a very large number of atomic emitters. Each atom radiates, at a specific instant, a **wave packet** (also known as a **wave train**) that lasts for about 10^{-8} s. Light radiated by a source is a mixture of wave packets emitted by different atoms at different instances (see Fig.20.3). Individual wave packets will be polarized, but each wave packet has its own polarization, and they are not correlated in any way. There is no continuity of plane of polarization and the plane of polarization varies from wave packet to wave packet in a completely random manner. The polarization of the waves determined at any particular spot will fluctuate randomly, and very rapidly, with no preferred direction. Any direction is equally likely, and the usual graphical representation of ordinary light is shown in the right-hand figure in Fig.20.4 (*a*). Fig.20.4 (*a*) is a fictitious diagram. It only implies that the natural light consists of electric field vectors of many possible orientations lying at different angles between 0 and 360° and hence symmetrically distributed about the direction of propagation.

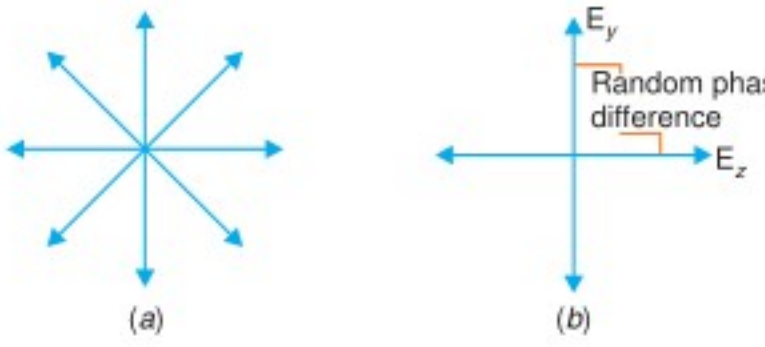


Fig.20.4: (*a*) Pictorial representation of Natural light. (*b*) Unpolarized light is viewed as a combination of *incoherent* vertically and horizontally polarized waves.

The concept of unpolarized light is rather difficult to visualize. In general, it is looked upon as consisting of an average of half of its vibrations horizontally polarized and half of its vibrations vertically polarized. Due to the random distribution of the optical vectors, the amplitude of the vertical and horizontal component vectors are taken as equal; however, the **two components are incoherent**, *i.e.* have a randomly changing phase difference. In view of this, unpolarized is pictorially represented by two electric vectors as shown in Fig.20.4 (*b*).

A comparison of Unpolarized and Polarized light

Unpolarized light	Polarized light
1. Consists of waves with planes of vibration equally distributed in all directions about the ray direction.	Consists of waves having their electric vector vibrating in a single plane normal to ray direction.
2. Symmetrical about the ray direction	Asymmetrical about the ray direction.
3. Produced by conventional light sources.	Is to be obtained from unpolarized light with the help of polarizers.
4. May be regarded as the resultant of two *incoherent* waves of equal intensity but polarized in mutually perpendicular planes.	May be regarded as the resultant of two mutually perpendicular *coherent* waves having zero phase difference.

20.5. TYPES OF POLARIZATION

The polarization of a light wave describes *the shape and locus of the tip of the E vector* (in the plane perpendicular to the direction of propagation) *at a given point in space as a function of time*. Depending upon the locus of the tip of the E vector, light may exhibit three different states of polarization. They are

(*i*) plane or linear polarization,

(*ii*) elliptical polarization and

(*iii*) circular polarization.

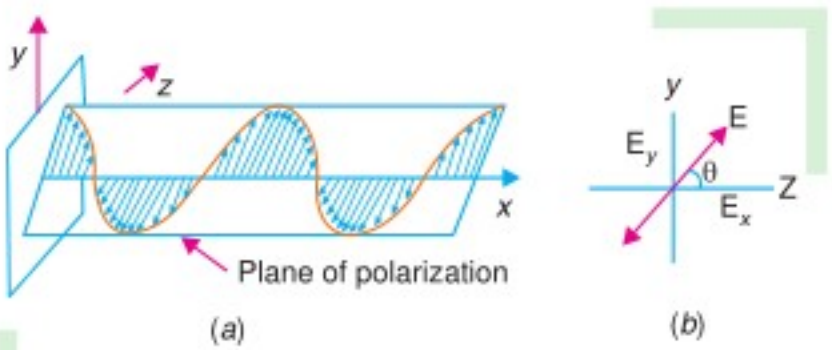
Apart from these, the light may also be partially polarized.

An unaided human eye cannot identify the state of polarization of light. However, some insects and animals possess polarization sensitive vision.

20.5.1 PLANE POLARIZED LIGHT

Plane polarized light waves are light waves in which the oscillations occur in a single plane. In a plane-polarized wave, the oscillations of electric field vector E are strictly confined to a single plane *perpendicular* to the direction of propagation. As the direction of the field vector at some point in space and time lies along a line in a plane perpendicular to the direction of wave propagation, a plane-polarized wave is also known as a **linearly polarized** wave.

With linear polarization, the orientation of the E-vector stays constant at a point in space. That is, the direction of E does not vary with time, but its magnitude varies sinusoidally with time. If the field is pointing either up or down, we call it **vertical polarization**, and if it's pointing either right or left, we call it **horizontal polarization**. Electric fields are not restricted to pointing exactly along vertical or horizontal axes, but can be at any arbitrary angle to those axes. Linearly polarized light, polarized at any arbitrary angle, may be regarded as a combination of horizontally and vertically polarized light, with appropriate amplitude, and which are oscillating **in phase** or **180° out**

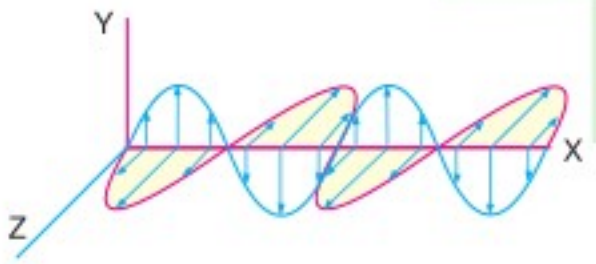


A light wave polarized in an arbitrary direction

Fig.20.5

of phase. The **key point** is that the two component waves are **coherent**.

Let ***x*** be the direction of travel of the light and, *y* and *z* be directions in the plane of the electric field. The electric field makes a constant angle to the *z*-direction, as shown in the Fig.20.5 (*a*). The wave in Fig.20.5 (*a*) is the resultant wave due to superposition of two **coherent** linearly polarized waves, oscillating in phase, as shown in Fig.20.6.

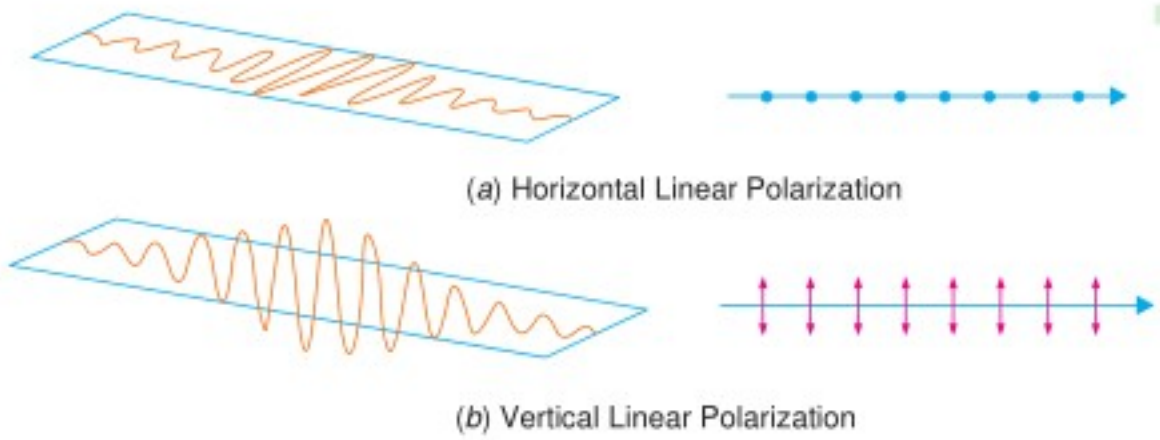


A linearly polarized wave is viewed as a combination of *coherent* vertically and horizontally polarized waves.

Fig.20.6

Representation of linearly polarized light in diagrams

Linearly polarized light is represented in diagrams as shown in Fig.20.7. When the electric field vector oscillates horizontally in a direction perpendicular to the plane of the paper, the light wave is represented by dots (Fig.20.7 *a*). When the electric field vector oscillates vertically in the plane of the paper, the light wave is represented by arrows, as shown in Fig.20.7 (*b*).



(*a*) Horizontal Linear Polarization

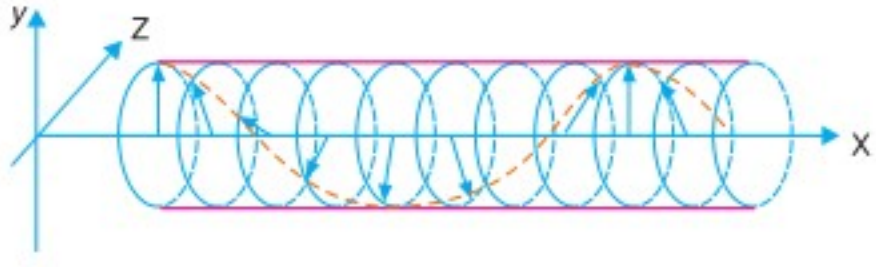
(*b*) Vertical Linear Polarization

Representation of linearly polarized light in diagrams

Fig. 20.7

20.5.2 CIRCULARLY POLARIZED LIGHT

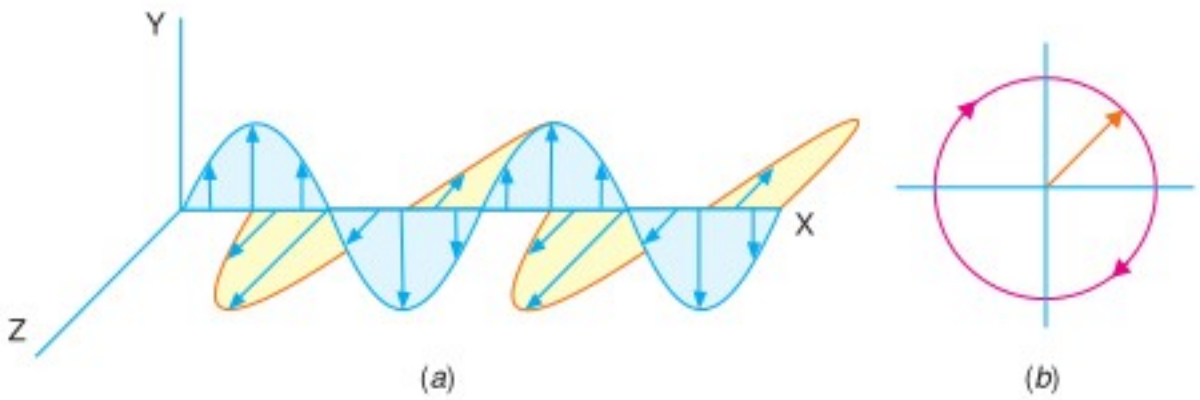
A light wave is said to be ***circularly polarized***, if in the course of wave propagation, the magnitude of the electric vector *E* stays constant but it rotates at a constant rate about the direction of propagation and sweeps a circular helix in space, as shown in Fig.20.8. This is a picture in terms of the **space variation** of *E*. Alternately, if we could see the wave advancing towards our eyes, we would find that the tip of the *E* vector tracing a circle in space, completing one revolution within one wavelength. Hence, the state of polarization is called *circular polarization*. In, circularly polarized light, there is no preference to specific direction of oscillation.



Space variation of *E*; *E* vector sweeps a circular helix in space

Fig. 20.8

A circularly polarized light wave may be regarded as the resultant wave produced due to superposition of two coherent linearly polarized waves of *equal amplitude* oscillating in mutually perpendicular planes, and are out of phase by 90°.

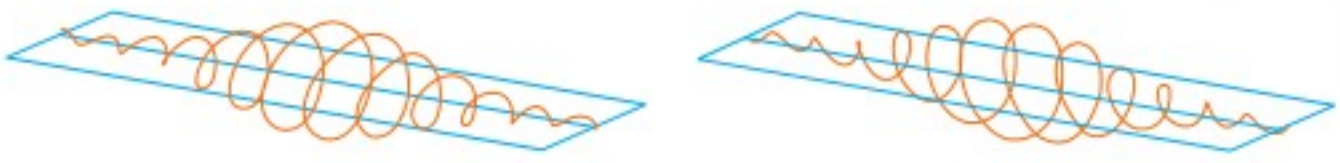


Circularly polarized wave is a combination of horizontally polarized and vertically polarized waves that are out of phase by 90° and having equal amplitudes.

Fig. 20.9

Let us again consider two linearly polarized waves having equal amplitude, out of which one is polarized in *z*-direction (horizontally polarized wave) and the other in *y*-direction (vertically polarized wave). Let us further assume that they are **coherent** and out of phase by 90° (see Fig.20.9a). At some instant, the *E*-field vector of *z*-polarized wave will have maximum amplitude and the *E*-field vector of y-polarized wave will be at zero. At that instant, the polarization is horizontal. A little time later, the *z*-polarized wave has decreased a little, while the *y*-polarized wave has started to increase. Then, the light looks like it is polarized at a slight angle— mostly horizontal, but with a small vertical component. A little while later, the *z*-component has decreased some more, and the *y*-component has increased some more, and the angle is greater. And so on. Eventually, the *y*-polarized wave is at a maximum, and the *z*-polarized wave is zero, and we have pure vertical polarization. If we stand at one point in space, and look at the direction of the wave, we will observe that the *E*-vector sweeps a circle in space. Hence, it is called circularly polarized wave. Note that the *oscillations of the resultant E-vector do not take place in a single plane.*

If the rotation of the tip of *E* is clockwise as seen by an observer looking back towards the source, then the wave is said to be **right-circularly polarized** (Fig.20.10 *a*). If the tip of *E* rotates anti-clockwise, then the wave is said to be **left-circularly polarized** (Fig.20.10. *b*). This is a description of circular polarization in terms of the **time variation** of *E*.



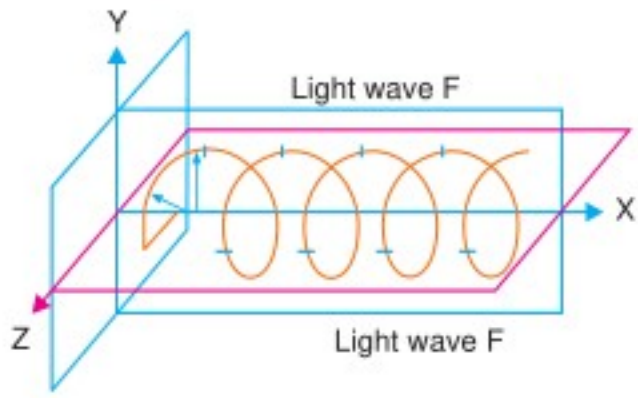
(a) Right circular polarization (b) Left circular polarization

Fig. 20.10

20.5.3 ELLIPTICALLY POLARIZED LIGHT

A light wave is said to be ***elliptically polarized***, if the magnitude of electric vector *E* changes with time and the vector *E* rotates about the direction of propagation and sweeps a flattened helix in space, as shown in Fig.20.11. This is a description of elliptically polarized light in terms of the **space variation** of ***E***. Alternately, if we imagine that we are looking at the light wave advancing towards us, we would observe that the tip of the *E* vector traces an ellipse in space. Hence, it is called elliptically

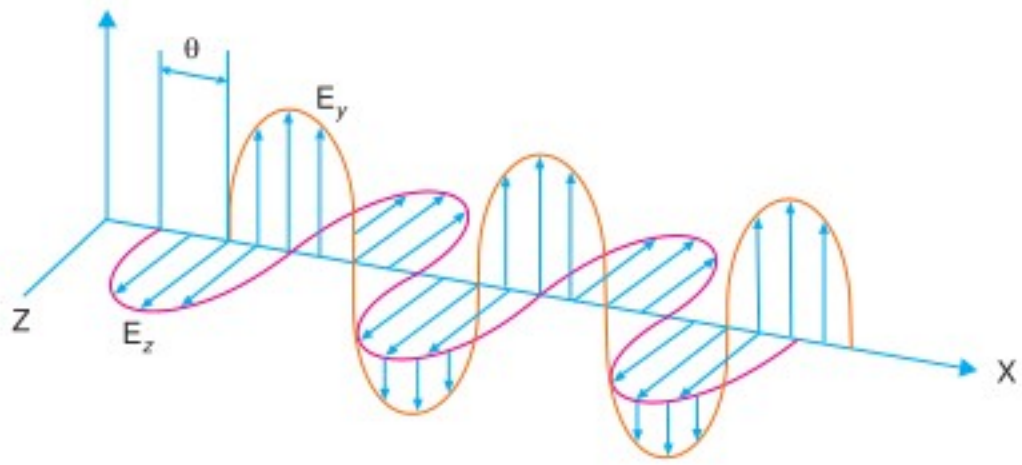
polarized light. This is a description of elliptical polarization in terms of the **time variation** of *E*.



Space variation of **E**; **E** vector sweeps a flattened helix in space

Fig. 20.11

An elliptically polarized light wave may be regarded as the resultant wave produced due to superposition of two **coherent** linearly polarized waves of different amplitudes, oscillating in mutually perpendicular planes and are out of phase. If waves of differing amplitude are related in phase by 90°, or if the relative phase difference is other than 90° then the resultant light wave is elliptically polarized.



Elliptically polarized wave is a combination of horizontally polarized and vertically polarized waves that are of different amplitudes and out of phase by θ

Fig. 20.12

Let us consider two linearly polarized waves having different amplitudes, out of which one is polarized in *x*-direction and the other in y-direction. Let us further assume that they are **coherent** and out of phase by an arbitrary angle θ (see Fig.20.12). The *E*-field vector of *Z*-polarized wave will have maximum amplitude at times when the *E*-field vector of *y*-polarized wave is a minimum and vice versa (Fig.20.12). Then the oscillations of the resultant *E*-vector do not take place in a single plane. The magnitude of resultant *E*-vector varies at each point in space and the overall rotation of the *E*-vector has the appearance of a flattened helix.

When we are looking back towards the source, if the rotation of *E* vector occurs clockwise, it is said to be a **right-elliptically-polarized** wave. If it rotates anti-clockwise, it is said to be a **left-elliptically polarized** wave.

20.5.4 PARTIALLY POLARIZED LIGHT

Usually, light is neither totally polarized nor unpolarized but a mixture of the two types. It can be viewed as a mixture of plane polarized light and unpolarized light. Partially polarized light is represented as shown in Fig.20.13.

Partially polarized light, like natural light, can be represented in the form of a superposition of two incoherent plane-polarized waves with mutually perpendicular planes of oscillations. In case of natural light the amplitude of these waves is the same and for partially polarized light, it is different.

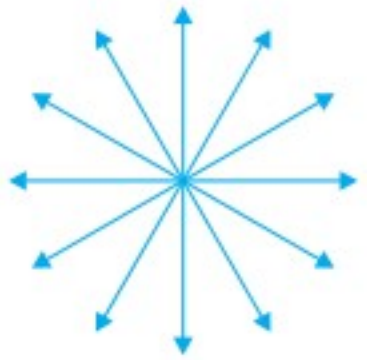


Fig. 20.13

Degree of Polarization:

If we pass partly polarized light through a polarizer, and if we rotate the polarizer about the direction of the ray, the intensity of the transmitted light will change within the limits from I_{max} to $I_{min.}$. The transition from one of these values to the other will occur upon rotation through an angle of 90°. We define the degree of polarization with the help of the following expression.

$$P = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} \quad \text{... (20.1)}$$

% polarisation

$$\% \text{ polarization} = \frac{(I_{max} - I_{min})}{(I_{max} + I_{min})} \times 100 \quad \text{... (20.2)}$$

For plane polarized light $I_{min} = 0$, and hence $P = 1$ and the % polarization is 100%. For natural light, $I_{max} = I_{min}$, and hence $P = 0$ and the % polarization is zero. If $I_{max} = 2I_{min}$, $P = 0.33$ and % polarisation = 100/3 = 33%.

The concept of the degree of polarization cannot be applied to elliptically and circularly polarized light.

20.6. PRODUCTION OF PLANE POLARIZED LIGHT

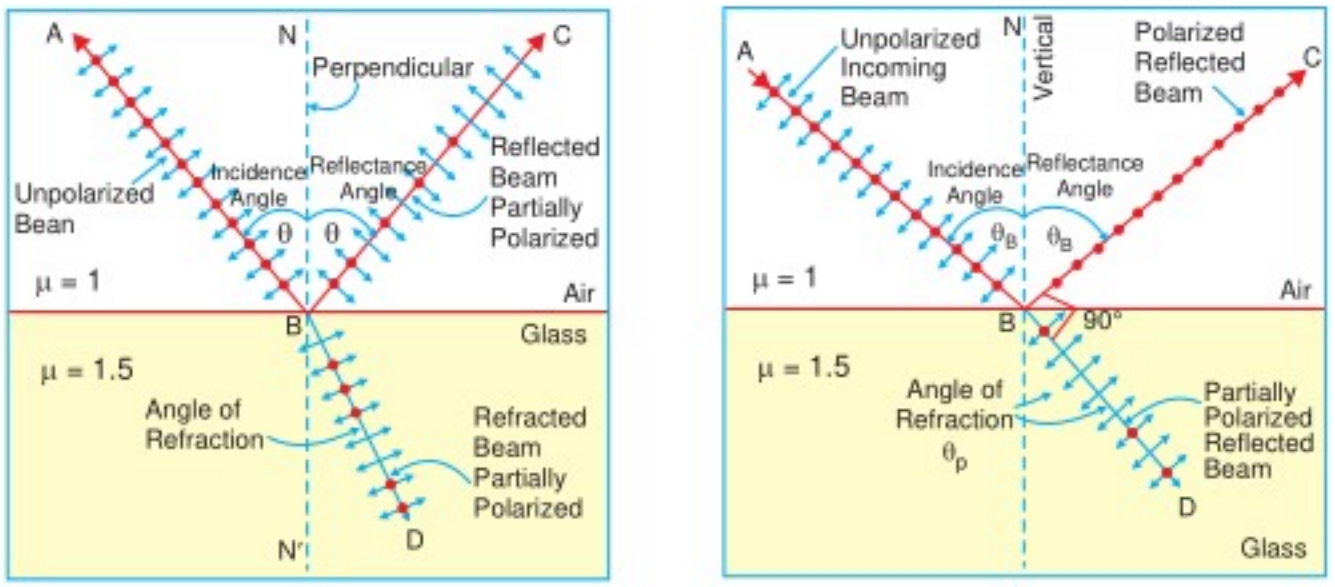
We now study the methods of producing plane-polarized light. Plane polarized light may be produced from unpolarized light using the following five optical phenomena:

(*i*) reflection, (*ii*) refraction, (*iii*) scattering, (*iv*) selective absorption (dichroism), and (*v*) double refraction. Out of these five, the phenomena of selective absorption and double refraction are helpful in practical production of plane polarized light.

20.6.1 POLARIZATION BY REFLECTION FROM DIELECTRIC SURFACES

E.L.Malus, French engineer discovered in 1808 polarization of light by reflection. He noticed that when natural light is incident on a smooth surface, at a certain angle the reflected beam is plane polarized. The extent to which polarization occurs is dependent upon the angle at which the light is incident on the surface and upon the material, which the surface is made of. Metallic surfaces reflect light with a variety of vibrational directions; such reflected light is unpolarized. However, light that is specularly reflected from dielectric surfaces, such as asphalt roadways, water etc, is linearly polarized. If the extent of linear polarization is large, a person perceives **glare** from such surfaces. On bright sunny days, the glare caused by sunlight on a roadway or a field of snow, may be almost blinding to the human eye.

When light wave is incident on a boundary between two dielectric materials, part of it is reflected, and part of it is transmitted.



(*a*) Reflection and Refraction at the surface between two media (*b*) Brewster Law

Fig. 20.14

Fig.20.14 (*a*) shows an unpolarized light beam AB incident on a glass surface. The incident ray AB and the normal NBN′ define the plane of incidence. The electric vector *E* of the ray AB can be resolved into two components, one perpendicular to the plane of incidence and the other lying in the plane of incidence. The perpendicular component is represented by **dots** and is called the ***s*-component**. The parallel component is represented by the **arrows** and is called the ***p*-component**. In case of completely unpolarized light the two components are of equal magnitude. At a particular angle θ_B, the reflection coefficient for *p*-component goes to zero and the reflected beam does not contain any *p*-component (see Fig.20.14 *b*). It contains only *s*-component and is totally *plane polarized*. The angle θ_B is called the **polarizing angle** or **Brewster's angle**.

This particular method of polarizing light is not advantageous, as the intensity of the reflected beam is very small.

20.6.1.1 Brewster's Law

Sir David Brewster performed a series of experiments on the polarization of light by reflection at a number of surfaces. He found that the polarizing angle depends upon the refractive index of the medium. In 1892, Brewster proved that **the tangent of the angle at which polarization is obtained by reflection is numerically equal to the refractive index of the medium.** If θ_B is the angle and m is the refractive index of the medium, then

$$\mu = \tan \theta_B \qquad \ldots (20.3)$$

This is known as Brewster's law.

If natural light is incident on a smooth surface at the polarizing angle, it is reflected along *BC* and refracted along *BD*, as shown in Fig.20.14 (*b*). Brewster found that the maximum polarization of reflected ray occurs when it is at right angles to the refracted ray. It means that $\theta_B + r = 90^o$.

$$\therefore \qquad r = 90^o - \theta_B \qquad \ldots (20.4)$$

According to Snell's law,

$$\frac{\sin \theta_B}{\sin r} = \frac{\mu_2}{\mu_1} \qquad \ldots (20.5)$$

where μ_2 is the absolute refractive index of reflecting surface and μ_1 is the refractive index of the surrounding medium. It follows from equ.(20.4) and equ.(20.5) that

$$\frac{\sin\theta_B}{\sin(90^\circ - \theta_B)} = \frac{\mu_2}{\mu_1}$$

or
$$\frac{\sin\theta_B}{\cos\theta_B} = \frac{\mu_2}{\mu_1}$$

$$\therefore \quad \tan\theta_B = \frac{\mu_2}{\mu_1} \qquad \ldots (20.6)$$

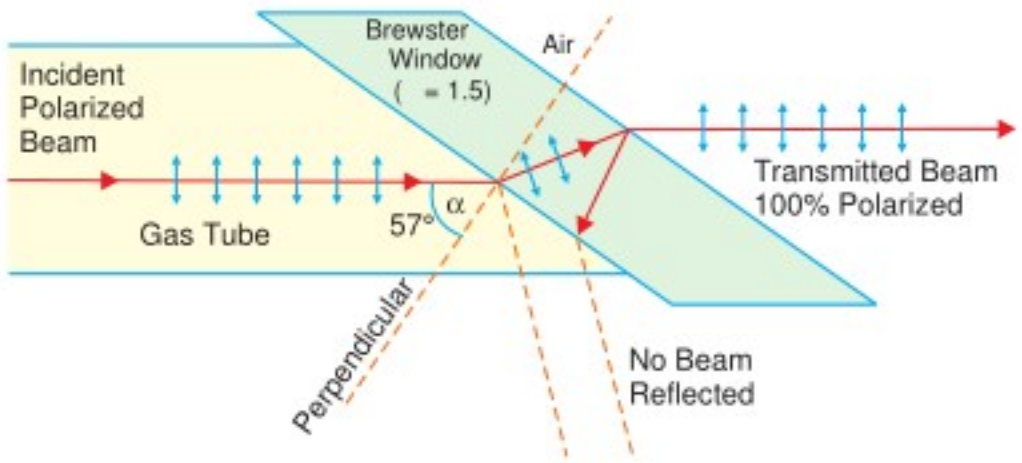
Equ.(20.6) shows that the polarizing angle depends on the refractive index of the reflecting surface. The polarizing angle q_B is known as **Brewster angle**. Light reflected from any angle other than Brewster angle is partially polarized.

Application of Brewster's law:

(*i*) Brewster's law can be used to determine the refractive indices of opaque materials.

(*ii*) It helps us in calculating the polarizing angle necessary for total polarization of reflected light for any material if its refractive index is known. However, the law is not applicable for metallic surfaces.

(*iii*) In gas lasers it is common to arrange two glass windows at the two ends of the laser tube. The windows are arranged at Brewster angle to the axis of the laser tube and hence they are called **Brewster windows**. The light beam traveling between the mirrors of the laser is reflected many times from these mirrors. Since the mirrors are at Brewster angle, all the light that is polarized perpendicular to the beam plane is emitted out of the laser cavity at early stage. In the gas tube remains only radiation polarized in the beam plane. The advantage of this arrangement is that the beam has no reflection losses, since only the transmitted polarized beam is traveling between the mirrors. The radiation out of these lasers is polarized as can be seen in Fig.20.15.



Brewster angle window at the end of gas laser

Fig. 20.15

(*iv*) Another application utilises the Brewster angle for transmitting a light beam into or out of an optical fibre without reflection losses.

20.6.2 POLARIZATION BY REFRACTION - PILE OF PLATES

When unpolarized light is incident at Brewster angle on a smooth glass surface, the reflected light is totally polarized, while the refracted light is partially polarized. If natural light is transmitted through a single plate, the transmitted beam is only partially polarized. If a stack of glass plates is

used instead of a single plate, reflections from successive surfaces occur leading to the filtering of the s-component in the transmitted ray. Ultimately, the transmitted ray consists of p-component alone. It is found that a stack of about 15 glass plates is required for this purpose. The glass plates are supported in a tube of suitable size and inclined at an angle of about 33° to the axis of the tube, as shown in Fig.20.16. Such an arrangement is called a **pile of plates**. Unpolarized light enters the tube and is incident on the plates at Brewster angle and the transmitted light will be totally polarized parallel to the plane of incidence.

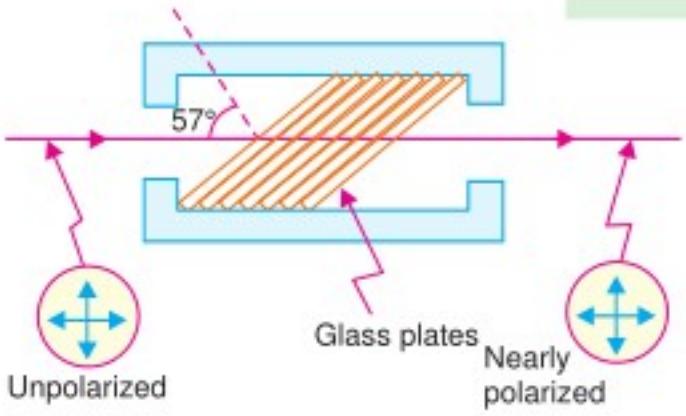


Fig. 20.16

20.6.3 POLARIZATION BY SCATTERING

If a narrow beam of natural light is incident on a transparent medium containing a suspension of ultramicroscopic particles, the light scattered is partially polarized. The incident light causes electrons in the scattering medium to vibrate. A vibrating electron emits most light in a direction perpendicular to its vibration and none along the direction of its vibration. The electric field of the emitted radiation is parallel to the direction of electron vibration. Hence light scattered through about 90° with respect to the incident direction is strongly polarized. The direction of vibration of E vector in the scattered light will be perpendicular to the plane defined by the direction of propagation and the direction of observation, *i.e.*, the plane of the paper, as illustrated in Fig.20.17.

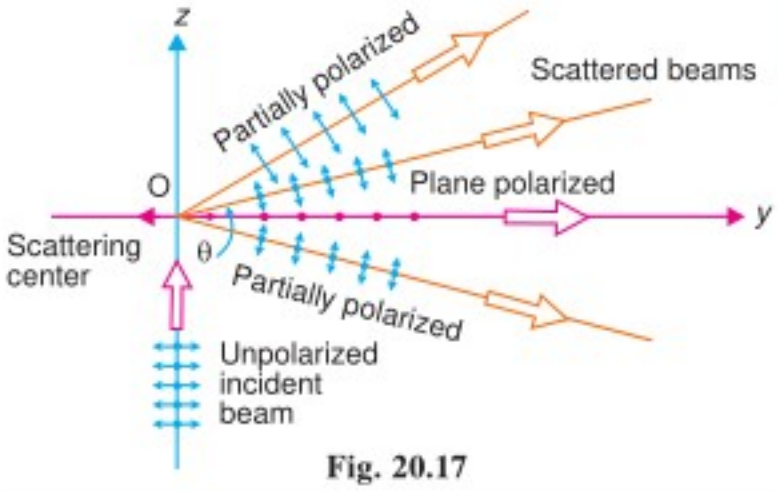


Fig. 20.17

The light from a blue sky is quite strongly polarised, particularly at 90° from the sun. It is not completely polarised because a significant amount of sunlight has undergone multiple-scattering, i.e. has been scattered more than once. Light scattered twice through a total angle of 90° would be less polarised than light scattered once.

20.6.4 POLARIZATION BY SELECTIVE ABSORPTION

A number of crystalline materials absorb more light in one incident plane than another, so that light progressing through the material become more and more polarized as they proceed. This difference in the absorption for the light rays is known as **selective absorption** or **dichroism**. Biot discovered this phenomenon in 1815. When natural light passes through a crystal such as tourmaline, it is split into two components, which are polarized in mutually perpendicular planes. The crystal absorbs light that is polarized in a direction parallel to a particular plane in the crystal but freely transmits the light component polarized in a direction perpendicular to that plane. If the crystal is of proper thickness, one of the components is totally absorbed and the other component emerging from the crystal is

linearly polarized. Selective absorption is illustrated in Fig.20.18.

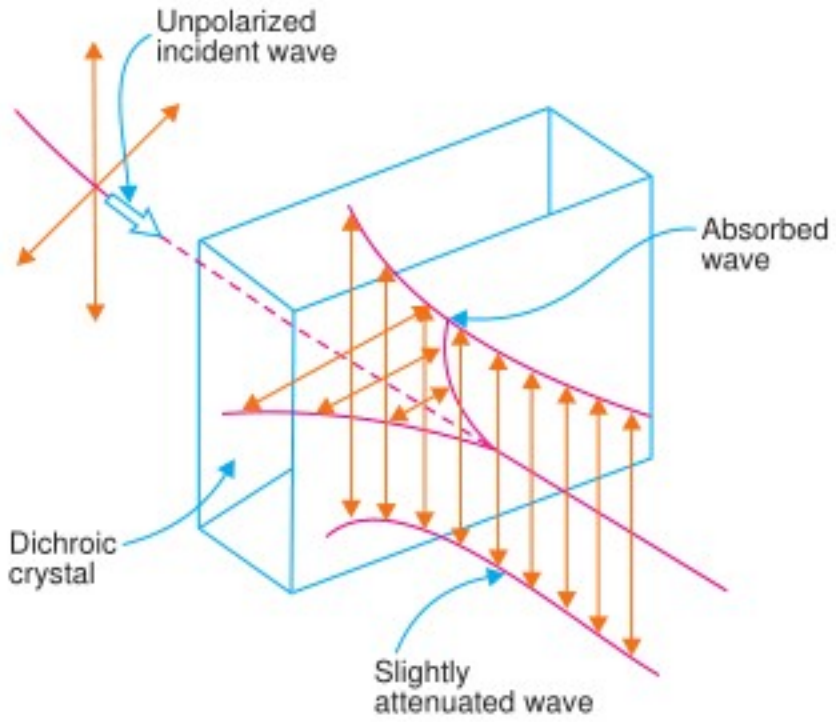


Fig. 20.18

20.6.5 POLARIZATION BY DOUBLE REFRACTION

When a beam of unpolarized light is incident on the surface of an anisotropic crystal such as calcite or quartz, it is found that it will separate into two rays (see Fig.20.19) that travel in different directions. This phenomenon is called **birefringence** or **double refraction**. The two rays are known as **ordinary ray (o-ray)** and **extraordinary ray (e-ray)**, which are linearly polarized in mutually perpendicular directions. A single linearly polarized ray is obtained in practice through elimination of one of the two polarized rays.

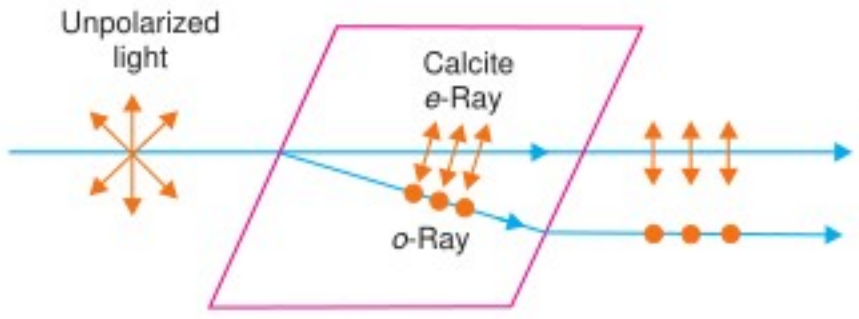


Fig. 20.19

20.7. POLAROID SHEETS

In 1928 E. H. Land invented Polaroid sheets, which utilize the phenomenon of selective absorption. The sheets are fabricated as follows. A clear plastic sheet of long chain molecules of PVA (polyvinyl alcohol) is heated and then stretched in a given direction to many times its original length. During the stretching process the PVA molecules become aligned along the direction of stretching. The sheet is then laminated to a rigid sheet of plastic to stabilise its size. It is then exposed to iodine vapour. The iodine atoms attach themselves to the straight long chain PVA molecules and consequently form long parallel conducting chains. The iodine atoms provide electrons, which can move easily along the aligned chains, but not perpendicular to them. When natural light is incident on the sheet, the electromagnetic vibrations that are in a direction parallel to the alignment of the iodine atoms are strongly absorbed because of the dissipative effects of the electron motion in the chains. Consequently,

only those vibrations in a direction perpendicular to the direction of molecular chains are transmitted. Thus, the light transmitted through the polaroid sheet is polarized. A sheet fabricated according to this process is known as H-sheet.

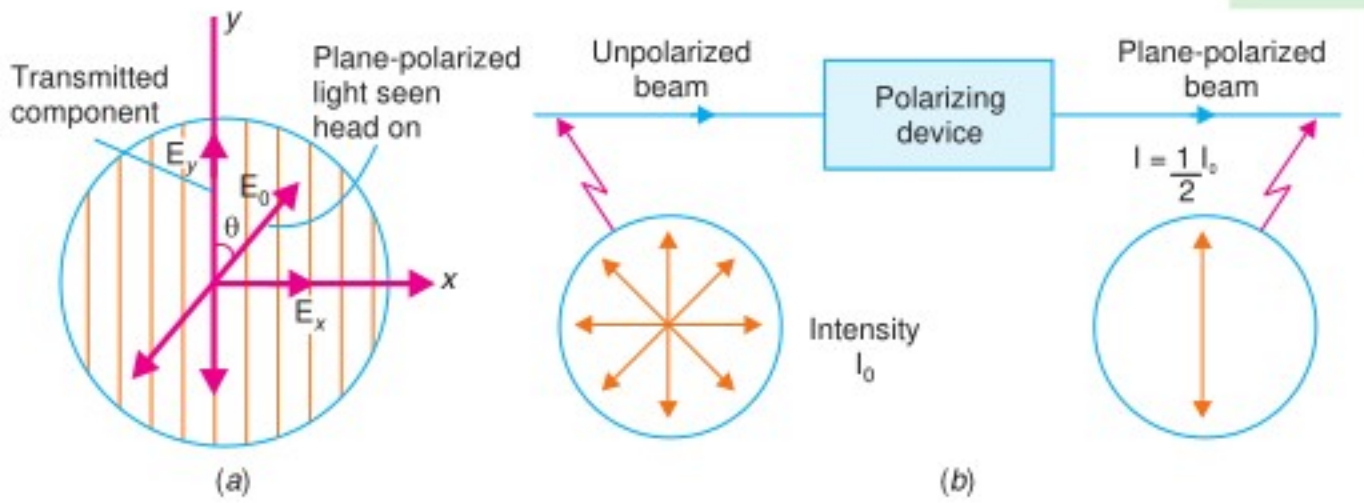
These sheet polarizers are inexpensive and can be made in large sizes. Polaroid sheets are widely used in sunglasses, camera filters etc to eliminate the unwanted glare from objects.

Polaroid sheets are extensively used for the production and detection of linearly polarized light.

20.8. POLARIZER AND ANALYZER

A *polarizer* is an optical element, which utilizes the phenomenon of selective absorption or double refraction, and transforms unpolarized light into polarized light. Plane polarized light is obtained by eliminating one of the two components in the unpolarized light. When natural light is incident on a polarizer, the *E*-field component that is parallel to the chains of iodine atoms induces current in the conducting chains and is therefore strongly absorbed. Consequently, the light transmitted contains only the component that is perpendicular to the direction of molecular chains. The direction of *E*-vector in the transmitted beam corresponds to the **transmission axis** of the Polaroid sheet.

Effect of polarizer on natural light



(*a*) Action of polarizer on linearly polarized wave
(*b*) The intensity of an unpolarized beam reduces to half after passing through a polarizer.

Fig. 20.20

Let us now understand the action of polarizer on the incident unpolarized light. Let us consider unpolarized light incident on a polarizer with the electric vector E_0 making an angle θ with respect to the *transmission axis* of the polarizer. The electric vector E_0 may be resolved into its component vectors lying parallel and perpendicular to the transmission axis of the polarizer (see Fig.20.20 *a*), that is E_y, parallel to the transmission axis and E_z, perpendicular to the transmission axis of the polarizer. The polarizer transmits the parallel component while blocking the perpendicular component. Thus, it is the parallel component E_y that is transmitted by the polarizer. But

$$E_y = E_0 \cos \theta \quad \ldots (20.7)$$

and hence, the intensity of the transmitted component is given by

$$I \propto E_y^2 = E_0^2 \cos^2 \theta \quad \ldots (20.8)$$

In unpolarized light all the values of θ are equally probable. Therefore, the fraction of light transmitted through the polarizer equals the average value of $\cos^2 \theta$.

$$\therefore \quad I = E_0(\cos^2\theta) = I_0(\cos^2\theta)$$

$$= \frac{I_0}{2\pi}\int_0^{2\pi}\cos^2\theta\, d\theta$$

$$= \frac{I_0}{2\pi}\int_0^{2\pi}\left(\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$= \frac{I_0}{4\pi}\left[(\theta)_0^{2\pi} + \left\{\frac{\sin 2\theta}{2}\right\}_0^{2\pi}\right]$$

$$= \frac{I_0}{4\pi}(2\pi + 0)$$

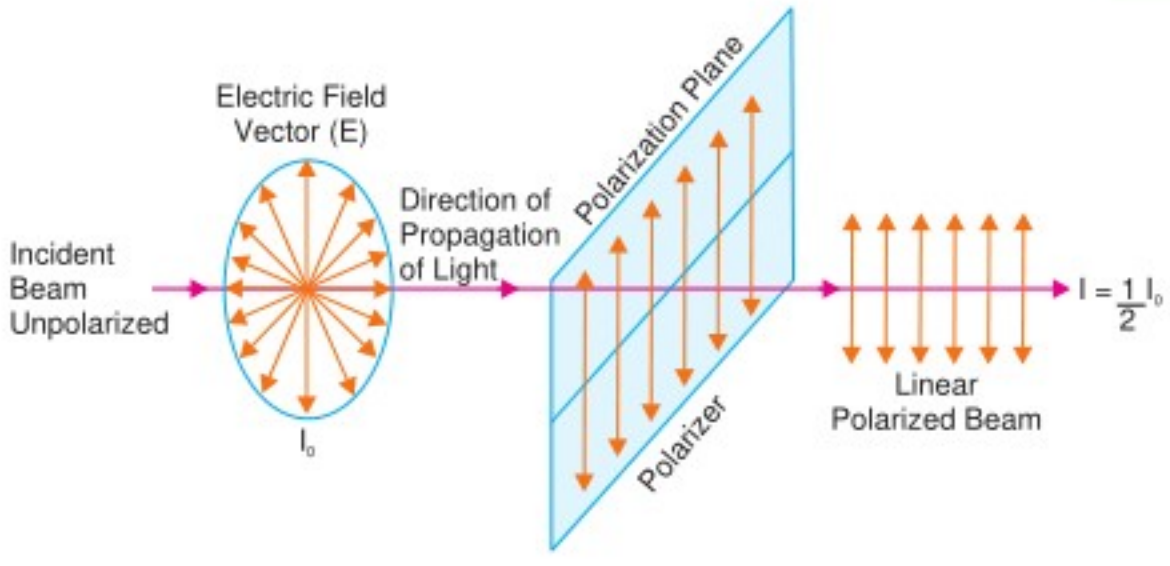
$$= \frac{1}{2}I_0$$

$$\therefore \quad I = \frac{1}{2}E_0^2 = \frac{1}{2}I_0 \qquad \ldots (20.9)$$

Thus, **if unpolarized light of intensity I_0 is incident on a polarizer, the intensity of light transmitted through the polarizer is $I_0/2$.**

20.8.1 PRODUCTION OF LINEARLY POLARIZED LIGHT USING A POLARIZER

A polarizer is associated with a specific direction called the **transmission axis** of the polarizer. If natural light is incident on a polarizer, only those vibrations that are *parallel* to the transmission axis are allowed through the polarizer whereas the vibrations that are in perpendicular directions are totally blocked. Therefore, the transmitted light contains waves oscillating in the same plane, as illustrated in Fig.20.21. Thus, the transmitted beam is linearly polarized.



Production of linearly polarized light

Fig.20.21

According to equ.(20.9), *when unpolarized light of intensity I_0 is incident on a polarizer, the intensity of light transmitted by the polarizer is $I_0/2$.*

20.8.2 DETECTION OF LINEARLY POLARIZED LIGHT

An *analyzer* is an optical element, which is used to identify the plane of vibration of plane polarized light.

To examine light coming from some direction either after emission or reflection etc, we use a Polaroid sheet. The Polaroid sheet used to determine the plane of polarization of light is known as an **analyzer**. There is no difference between a polarizer and analyzer in fabrication but they differ in their roles. Both the polarizer and analyzer are characterized by a transmission axis.

When the transmission axis of the analyzer A is set up parallel to that of polarizer P, light transmitted by the polarizer, passes unhindered through the analyzer (Fig. 20.22 *a*).

If the transmission axes are set at an angle q, light is partially transmitted (Fig.20.22 *b*). As the angle rotates from 0 to 90 degrees, the amount of light that is transmitted decreases. When the axes are perpendicular to each other, the polarized light from P is extinguished by the analyzer A (Fig.20.22 *c*). The polarizer and analyzer are said to be **crossed** in this configuration.

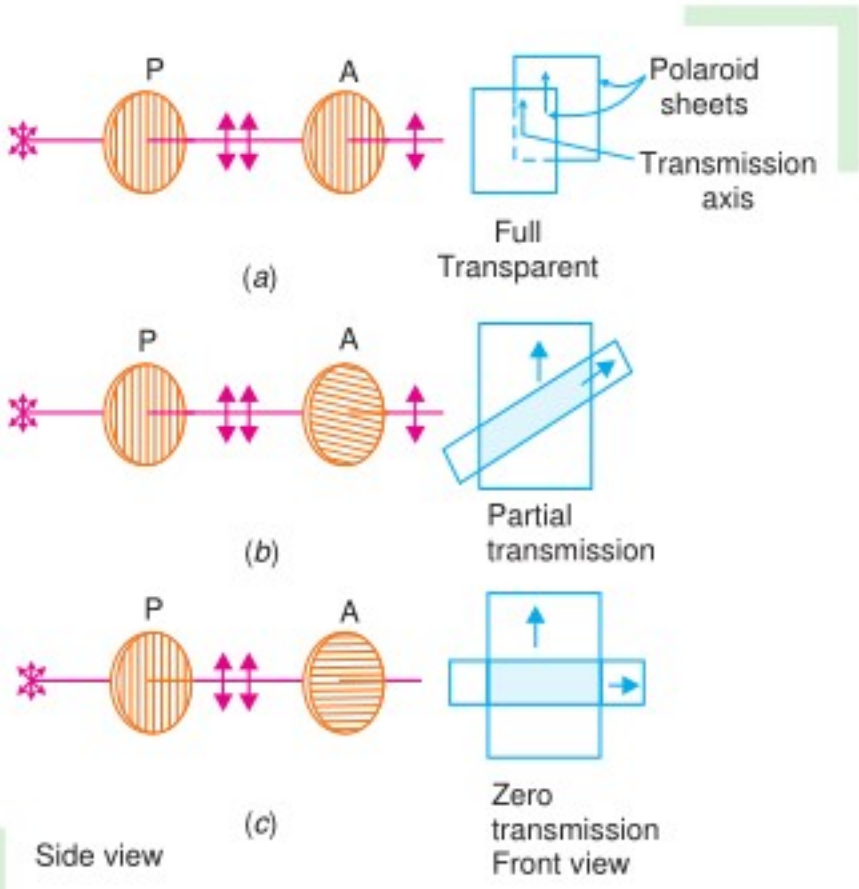


Fig. 20.22

When we rotate the axis of the analyzer with respect to that of the polarizer, we obtain two positions of maximum intensity and two positions of zero intensity in *one full rotation*. Conversely, if we obtain two positions of maximum intensity and two positions of zero intensity in one full rotation of the analyzer, we conclude that the light incident on the analyzer is plane-polarized light.

20.9. MALUS' LAW

The amount of light transmitted through a polarizer at an arbitrary angle [Fig.20.22 (*b*)] is given by **Malus's Law**.

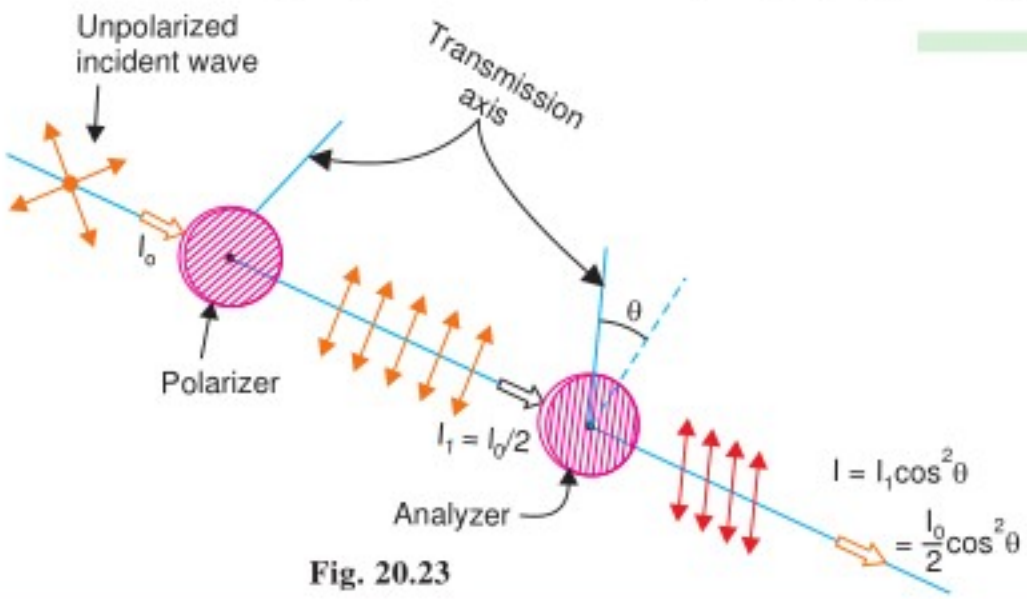


Fig. 20.23

In 1809 Malus found that the *intensity of polarized light transmitted through a polarizer is proportional to the square of cosine of the angle between the plane of polarization of the light and the transmisson axis of the polarizer.*

This statement is known as Malus' law.

If unpolarized light of intensity I_o is incident on a polarizer, plane polarized light of intensity $I_o/2$ is transmitted by it. Let us denote $I_o/2$ by I_1. Let this plane polarized light pass through an analyzer. The intensity of the light transmitted through the analyzer is given by

$$I = E_1^2 \cos^2\theta = I_1 \cos^2\theta = \frac{1}{2} I_0 \cos^2\theta \quad \ldots (20.10)$$

Light transmitted through the analyzer at specific settings are as follows.

Case (*i*): If $\theta = 0°$	axes parallel	$I = I_1 = \frac{1}{2} I_0$
Case (*ii*): If $\theta = 90°$	axes perpendicular	$I = 0$
Case (*i*): If $\theta = 180°$	axes parallel	$I = I_1 = \frac{1}{2} I_0$
Case (*i*): If $\theta = 270°$	axes perpendicular	$I = 0$

Thus, we obtain two positions of maximum intensity and two positions of zero intensity when we rotate the axis of the analyzer with respect to that of the polarizer.

20.10. ANISOTROPIC CRYSTALS

When a light beam is incident on an *isotropic medium* such as a glass slab, it refracts as a single ray. An optically isotropic material is one in which the index of refraction is the same in all directions. Glass, water and air are examples of isotropic materials. The atoms in a crystal are arranged in a regular periodic manner. If the arrangement of atoms differ in different directions within a crystal, then the physical properties vary with the direction. The thermal conductivity, electrical conductivity, velocity of light and hence refractive index etc properties depend on the crystallographic direction along which the property is measured. Then we say that the crystal is **anisotropic**. In such anisotropic crystals the force of interaction between the electron cloud and the lattice is different in different crystallographic directions. The natural frequency of the electron cloud is likewise dependent on the direction in which the electrons are caused to vibrate by the incident light wave. This results in different velocities in different directions and the index of refraction is different in different directions within the crystal.

The anisotropic crystals are divided into two classes: uniaxial and biaxial crystals. In case of **uniaxial crystals**, one of the refracted rays is an ordinary ray and the other is an extraordinary ray. In **biaxial crystals** both the refracted rays are extraordinary rays. Calcite, tourmaline and quartz are examples of uniaxial crystals whereas mica, topaz and aragonite are examples of biaxial crystals.

20.10.1 CALCITE CRYSTAL

Calcite is a common naturally occurring substance. Both marble and limestone are made up of many small calcite crystals bonded together. A large crystal of calcite is colourless and transparent. It was at one time found in great quantities in Iceland and hence it is also known as **Iceland spar**. Naturally occurring calcite crystals (Fig.20.24*a*) has rhombohedral cleavage which means it breaks into blocks with parallelogram-shaped faces. It is bounded by six faces (Fig.20.24*b*), each of which is a parallelogram with angles equal to 101°55'and 78°5'. The rhombohedron has only two corners A and H where all the face angles are obtuse (101°55'). These two corners appear as the *blunt corners* of the crystal. At the rest of six corners there is one obtuse angle and two acute angles.

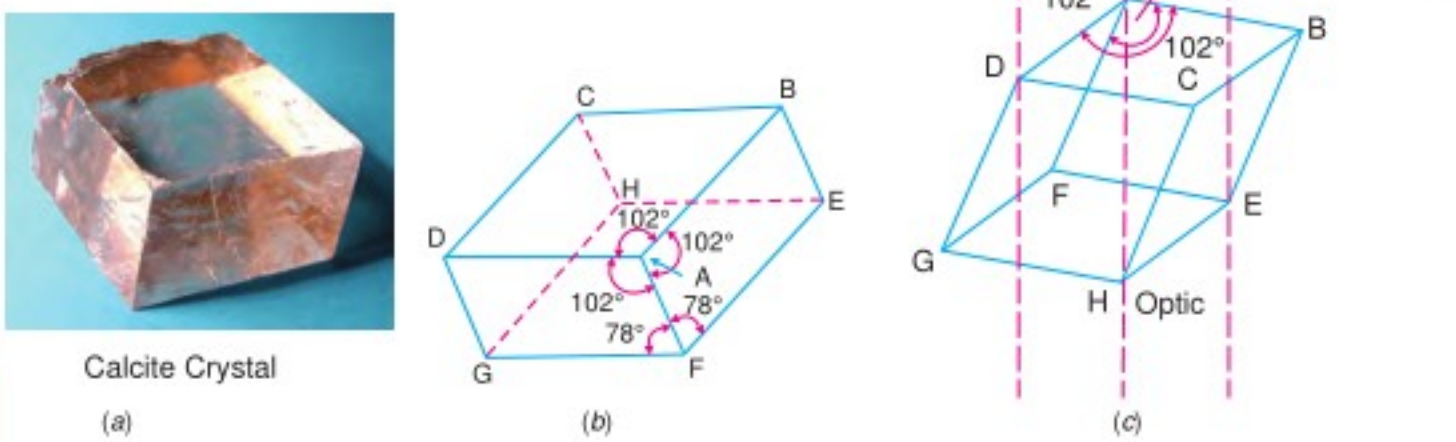


Fig. 20.24

20.10.2 OPTIC AXIS

A line bisecting any one of the blunt corners (A or H) and making equal angles with each of the three edges meeting there, is the **optic axis** (see Fig.20.24 *c*). In fact any line parallel to this line is also an optic axis. Thus, the optic axis is a direction and not a specific line in the crystal. Hence an optic axis can be drawn through every point in the crystal, that is, any line parallel to the line above will also be the optic axis. It is to be noted that the optic axis is not obtained by joining the two blunt corners. Only in a special case, when the three edges of the crystal are equal, the line joining the two blunt corners A and H coincide with the crystallographic axis of the crystal and it gives the direction of the optic axis. The optic axis is actually the axis of symmetry of the crystal. A ray of light propagating along optic axis does not suffer double refraction, because the structure of the crystal is symmetric about that direction.

The optic axis is the direction in a uniaxial crystal along which the e-ray and the o-ray travel with the *same* speed and consequently double refraction *does not* take place along this direction. The corresponding refractive index is the refractive index for ordinary light, say μ_o.

20.10.3 PRINCIPAL SECTION

A plane containing the optic axis and perpendicular to a pair of opposite faces of the crystal is called the **principal section** of the crystal for that pair of faces (Fig.20.25 *a*). Thus, there are three principal sections passing through any point within the crystal, one corresponding to each pair of opposite faces. A principal section always cuts surfaces of calcite crystal in a parallelogram having angles 71° and 109° (see Fig.20.25 *b*). Fig.20.25 (*c*) shows a face of the crystal in which the end-view

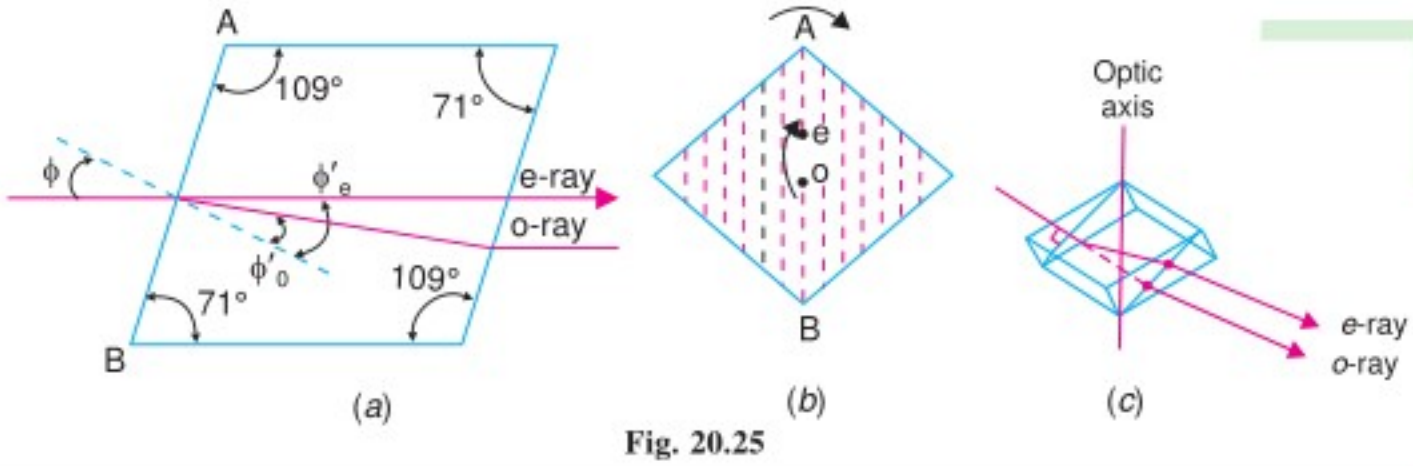


Fig. 20.25

of the principal section AB (of Fig.20.25 b) is shown by the dotted line AB. The lines parallel to AB represent the end-views of other principal sections parallel to AB with in the crystal.

20.10.4 PRINCIPAL PLANE

Defining principal section is not enough to understand the directions of vibrations for the o-ray and e-rays. Hence, two more planes are defined as principal plane for the o-ray and the principal plane for the e-ray. The plane containing the optic axis and the o-ray is called the *principal plane of the o-ray* and the plane containing the optic axis and the e-ray is called the *principal plane of the e-ray*. The directions of vibrations in the o-ray and e-ray can be understood with reference to these planes. In general, the two principal planes do not coincide. Under the particular case, when the plane of incidence is the principal section of the crystal, then the principal planes of o- and e-rays and the principal section of the crystal coincide.

20.11. DOUBLE REFRACTION IN CALCITE CRYSTAL

Fig.20.25 (*a*) shows a principal section of calcite crystal. A ray of light is incident on the face AB of the crystal and it travels along the principal section. The ray is split into two rays, namely o-ray (fast ray) and e-ray (slow ray). The o-ray travels through the crystal without deviation while the e-ray is refracted at some angle. As the opposite faces of the crystal are parallel, the rays emerge out parallel to the incident ray. Within the crystal the o-ray *always* lies in the plane of incidence whereas e-ray does not lie in the plane of incidence. e- ray lies in the plane of incidence only when the plane of incidence is a principal section.

If a mark (dot or cross) is made on a paper and then the calcite crystal (AB face) is placed on it, two images are seen through the crystal, as illustrated in Fig.20.25 (*b*). The images are produced by the o-ray and e-ray. The intensities of the images are lesser than that of the original mark. The line joining them lies in the principal section. If now the crystal is rotated slowly about an axis passing through the o-image, the e-image moves round in a circle while the o-image remains stationary. It shows that the velocity of propagation of o-ray is the same in all directions, while that of e-ray changes with direction. O-ray obeys the laws of refraction and the e-ray does not follow the ordinary laws of refraction.

The e-ray and o-ray are linearly polarized. The e-ray has its vibrations (i.e., the optical vector) *parallel* to the principal section whereas the vibrations (optical vector) in o-ray are *perpendicular* to the principal section, as indicated in Fig.20.19. The vibration directions can be established by examining the rays through a polarizer. As the polarizer is held in the path of the rays and rotated slowly, the intensity of one of the images, say the o-image, increase while that of the e-image decreases. In one position, the intensity of the o-image will be a maximum while the e-image is extinguished. Further rotation through 90° from this particular position, causes the o-image to disappear and e-image intensity to become a maximum. It proves that the e- and o-rays are linearly polarized in mutually perpendicular directions.

The o-ray travels with the same velocity in all directions in the crystal whereas the e-ray travels with different velocities in different directions. Therefore, refractive index corresponding to o-ray is a constant and is designated by μ_o. The refractive index corresponding to e-ray varies and is denoted by μ_e. The difference between the refractive indices is known as the amount of double refraction or birefringence. Thus,

$$\Delta_\mu = \mu_e - \mu_o \qquad \ldots (20.11)$$

20.11.1 HUYGENS' EXPLANATION OF DOUBLE REFRACTION

In order to explain the double refraction exhibited by anisotropic crystals, Huygens postulated that the incident light excites two separate wavelets within the crystal, one spherical wavelet associated

with the ordinary waves and one ellipsoidal wavelet associated with the extraordinary waves. For example, the plane wavefront, in Fig.20.26, incident normally on the crystal surface generates spherical wavelets as well as ellipsoidal wavelets. The spherical wavelets propagate equally quickly in all directions. The wave surface corresponding to o-ray is therefore spherical. The tangent to these waves lies straight ahead and, by successive application of the principle, the plane wave propagates straight ahead with speed υ. The ellipsoidal wavelets propagate at different speeds in different directions. The wave surface corresponding to e-ray is therefore an ellipsoid of revolution about the optic axis. The common tangent to these ellipsoids after a little time is the new wavefront. The line from the point of generation of each ellipsoid to the tangent point on that ellipsoid is off at an angle and defines the direction of travel of the extraordinary wavefronts. The wavefronts are not perpendicular to their direction of travel.

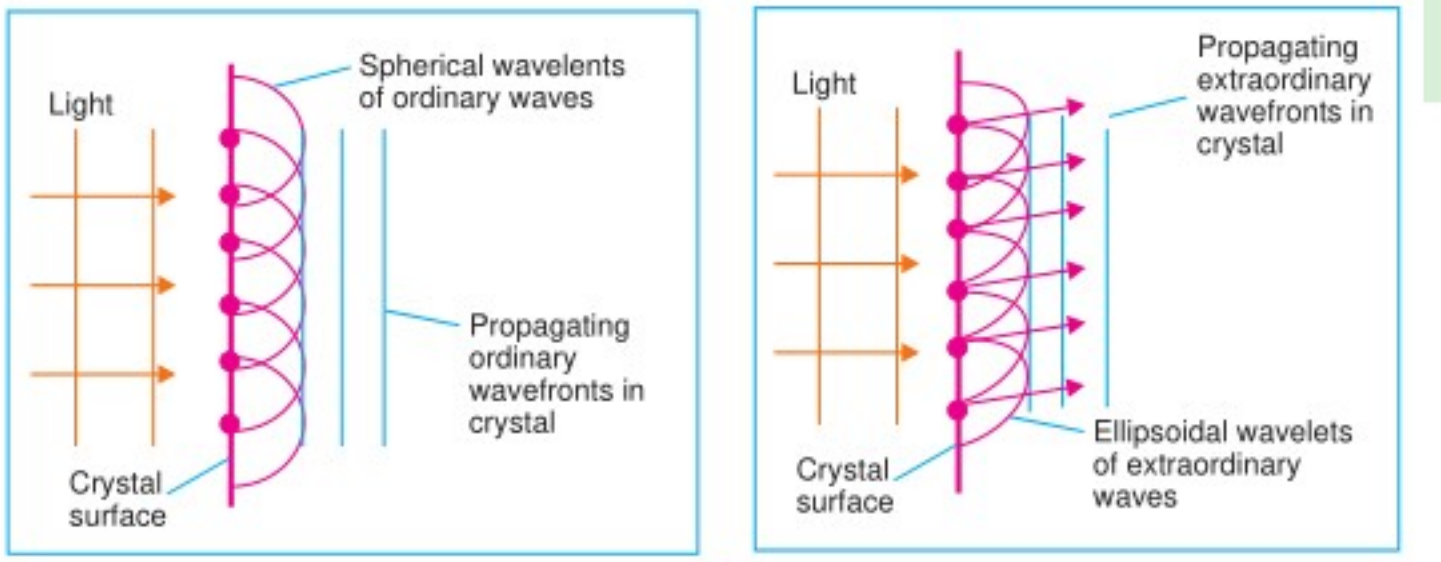


Fig. 20.26

The two wave surfaces touch each other at the two points where they are intersected by the optic axis. As light propagates through the crystal, the two wave surfaces travel in different directions in the crystal. Ultimately, two refracted rays emerge from the crystal.

20.11.2 ORDINARY AND EXTRA-ORDINARY RAYS

We now compare the properties of o- and e-rays:

(*i*) o-ray obeys the laws of refraction and the e-ray does not follow the ordinary laws of refraction.

(*ii*) Both o-ray and e-ray are plane polarized. They are polarized in mutually perpendicular planes. The electric vector of o-ray vibrates perpendicular to the principal section of o-ray while the vibrations of e-ray take place parallel to the principal section of e-ray.

(*iii*) O-ray travels with the same speed in all directions within the crystal. The e-ray travels with different speeds along different directions in the crystal. However, the speed of e-ray will be equal to that of o-ray along the optic axis direction.

(*iv*) Because o-ray travels with the same velocity in all directions, the refractive index corresponding to it has a constant value. On the other hand, the refractive index for e-ray varies from direction to direction. The principal refractive index for o-ray is defined as follows:

$$\mu_0 = \frac{c}{\upsilon_0} = \frac{\text{velocity of light in a vacuum}}{\text{velocity of o-ray in the crystal}} \qquad \ldots (20.12)$$

The principal refractive index for e-ray in ***positive crystals*** is defined as follows:

$$\mu_o = \frac{c}{(\upsilon_e)_{min}} = \frac{\text{velocity of light in a vacuum}}{\text{minimum velocity of e-ray in the crystal}} \quad \ldots (20.13)$$

The principal refractive index for e-ray in ***negative crystals*** is defined as follows:

$$\mu_e = \frac{c}{(\upsilon_e)_{max}} = \frac{\text{velocity of light in a vacuum}}{\text{maximum velocity of e-ray in the crystal}} \quad \ldots (20.14)$$

(*v*) When natural light is incident on an anisotropic crystal at an angle to the optic axis, it splits into o-and e-rays, which travel in *different directions* with *different velocities* (20.27 *a*).

When natural light is incident in a direction perpendicular to the optic axis, o-ray and e-ray propagate in the *same direction* in the crystal but with *different velocities*, as shown in Fig.20.27 (*b*). In a negative crystal e-ray leads o-ray and in case of a positive crystal o-ray leads e-ray.

When natural light is incident on the crystal in a direction parallel to the optic axis, it does not split into two rays. The o- and e- rays travel in the *same direction with the same velocity*, as shown in Fig.20.27 (*c*).

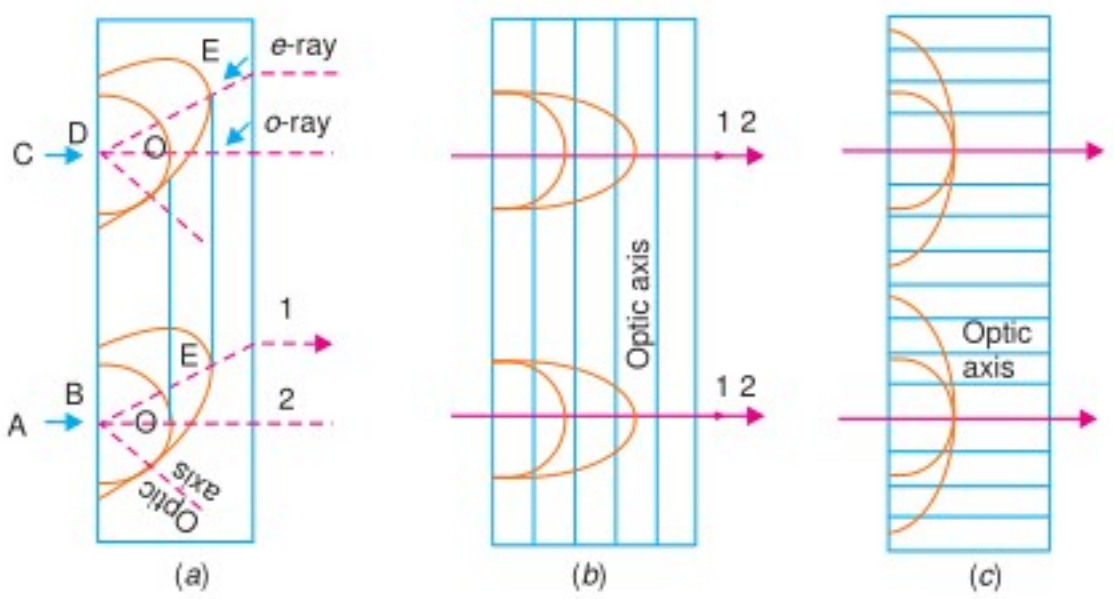


Fig. 20.27

(*vi*) The distinction of o-ray and e-ray exists only within the crystal. Once they emerge from the crystal, they travel with the same velocity. The rays outside the crystal differ only in their direction of travel and plane of polarization. The designation of o- ray and e-ray has no meaning outside the crystal.

20.11.3 POSITIVE AND NEGATIVE CRYSTALS

Because of two different wave fronts, two different types of uniaxial crystals exist. In one type of crystals, the spherical wave front of o-ray is enclosed by the ellipsoidal wave front of e-ray. Such crystals are known as **negative crystals**. They are called negative crystals because the refractive index corresponding to the e-ray is less than that corresponding to o-ray. Calcite crystal is an example of negative type crystals. In the other case, the extraordinary wave front lies within the ordinary wave front and such crystals are called **positive crystals**. They are positive because the refractive index for the extraordinary ray is greater than that of o-ray. Quartz crystal is an example of positive crystals.

We compare here the characteristics of the positive and negative crystals.

(*i*) In positive uniaxial crystals, the ellipsoid of revolution corrrsponding to the e-ray is totally contained within the sphere corresponding to the o-ray.

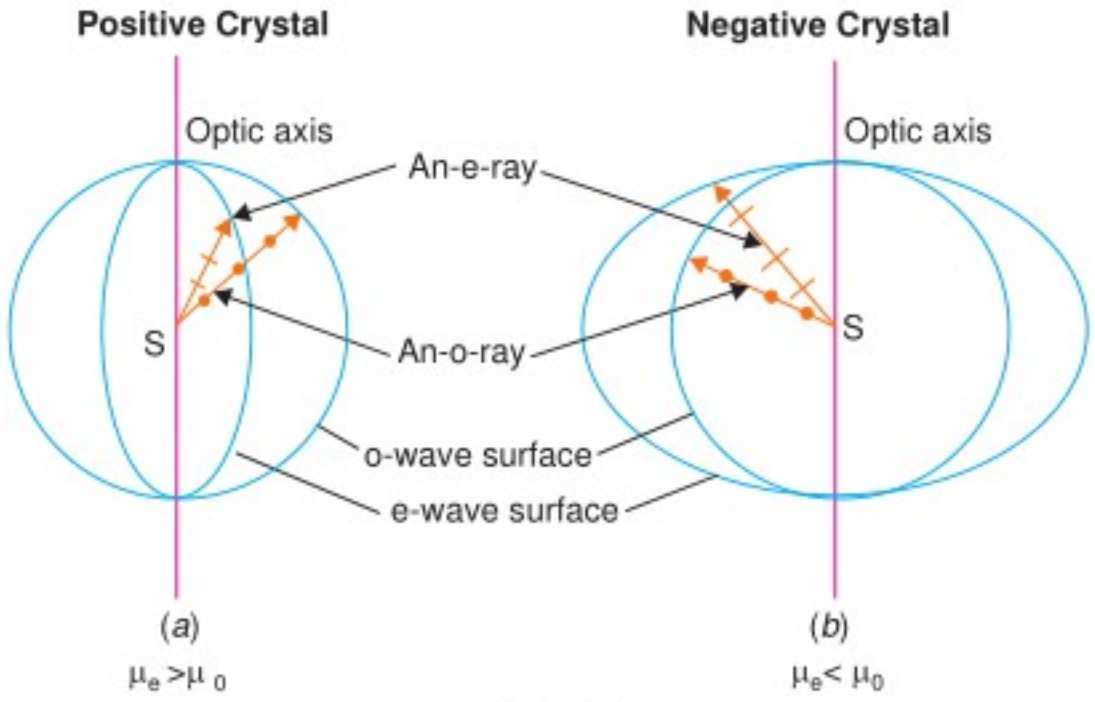


Fig. 20.28

In negative uniaxial crystals, the ellipsoid of revolution for e-ray lies completely outside the sphere corresponding to o-ray. The two cases are depicted in Fig. 20.28.

(*ii*) In positive crystals the e-ray velocity has a maximum value along the optic axis and a minimum value in a direction perpendicular to the optic axis.

On the other hand, in negative crystals the velocity of e-ray has a minimum value parallel to the optic axis and a maximum value in a direction perpendicular to the optic axis.

(*iii*) In positive crystals, e-ray travels slower than o-ray in all directions except along the optic axis.

$$v_e = v_o \quad — \quad \text{parallel to optic axis}$$
$$v_e < v_o \quad — \quad \text{other directions}$$

In negative crystals, o-ray travels slower than e-ray in all directions except along the optic axis.

$$v_e = v_o \quad — \quad \text{parallel to optic axis}$$
$$v_e > v_o \quad — \quad \text{other directions}$$

(*iv*) In positive crystals the principal refractive index for e-ray is larger than the principal refractive index for o-ray.

$$\mu_e > \mu_o$$

In negative crystals the principal refractive index for o-ray is larger than the principal refractive index for e-ray.

$$\mu_e < \mu_o$$

(*v*) **Birefringence** or amount of double refraction of a crystal is defined as

$$\Delta\mu = \mu_e - \mu_o \qquad \ldots (20.15\ a)$$

As $\mu_e > \mu_o$ in these crystals, $\Delta\mu$ is a positive quantity for positive crystals.

$\Delta\mu$ is a negative quantity for negative crystals as $\mu_e < \mu_o$ in these crystals and the birefringence of a negative crystal is given by

$$\Delta\mu = \mu_o - \mu_e \qquad \ldots (20.15\ b)$$

20.12. NICOL PRISM

Nicol prism is a polarizing device fabricated from a double refracting crystal. It is similar to a Polaroid sheet in its action. A Nicol prism is made from calcite crystal. William Nicol designed it in 1820. A rhomb of calcite crystal about three times as long as it is thick, is obtained by cleavage from the original crystal. The ends of the rhombohedron are ground until they make an angle of 68°instead of 71° with the longitudinal edges. This piece is then cut into two along a plane perpendicular both to the principal axis and to the new end surfaces *AB* and *CD*. The two parts of the crystal are then cemented together with canada balsam, whose refractive index lies between the refractive indices of calcite for the o-ray and e-ray. μ_o=1.66, μ_e=1.486 and $\mu_{\text{canada balsam}}$=1.55. The position of the optic axis is as shown in Fig.20.29. The refractive index for e-ray depends upon the direction in which e-ray is propagating in the crystal. The difference between the refractive index between o-ray and that for e-ray goes on increasing with the angle between the two rays in the crystal. When this angle is 90°, the difference is a maximum. Thus, for a fixed value for μ_o, the μ_e has its maximum or minimum value in perpendicular direction. In the above μ_e = 1.486 represents the minimum value.

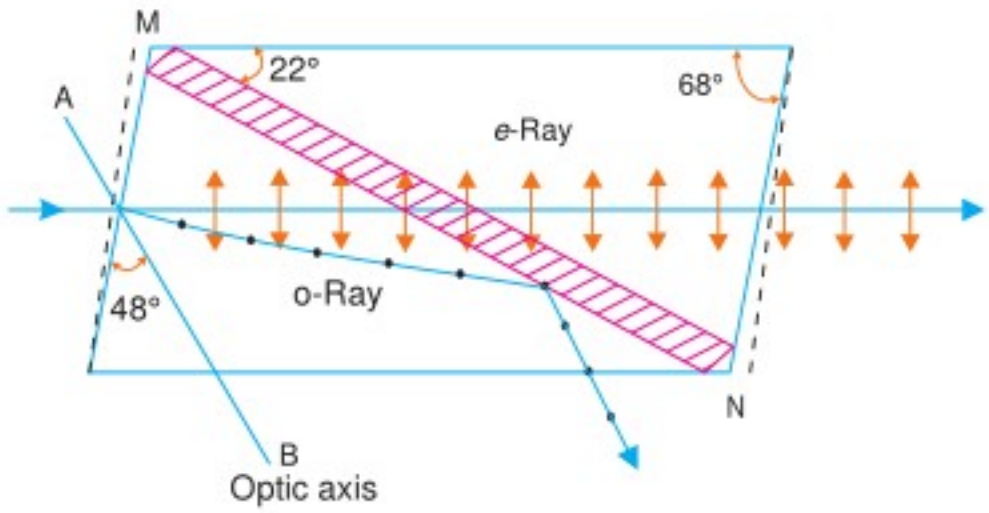


Fig. 20.29

Unpolarized light is made to fall on the crystal as shown in Fig.20.29 at an angle of about 15°. The ray after entering the crystal suffers double refraction and splits up into o-ray and e-ray. The two rays with their directions of vibrations are as shown in the Fig.20.29. The values of the refractive indices and the angles of incidence at the canada balsam layer are such that the e-ray is transmitted while the o-ray is internally reflected. The face where the o-ray is incident is blackened so that the o-ray is completely absorbed. Then we get only the plane-polarized e-ray coming out of the Nicol. Thus, the Nicol works as a polarizer.

For studying the optical properties of transparent substances, two Nicols are used - one as a polarizer and the other as an analyzer.

When two Nicol prisms P and A are placed adjacent to each other as shown in Fig.20.30, one of them acts as a polarizer and the other acts as an analyser. If unpolarized ray of light is incident on the Nicol prism P, a linearly polarized e-ray emerges from P with its vibration direction lying in the principal section of P. The state of the polarization of the light emerging from polarizer P can be examined with another polarizer A, which for convenience is called an analyser. Let now this ray be incident on the second Nicol prism A, whose principal section is parallel to that to P. The vibration direction of the ray will be in the principal section of A and hence it is transmitted unhindered through the analyser A.

If the Nicol prism A is gradually rotated, the intensity of the e-ray decreases in accordance with Malus law. When its principal section becomes perpendicular to that of the Nicol prism

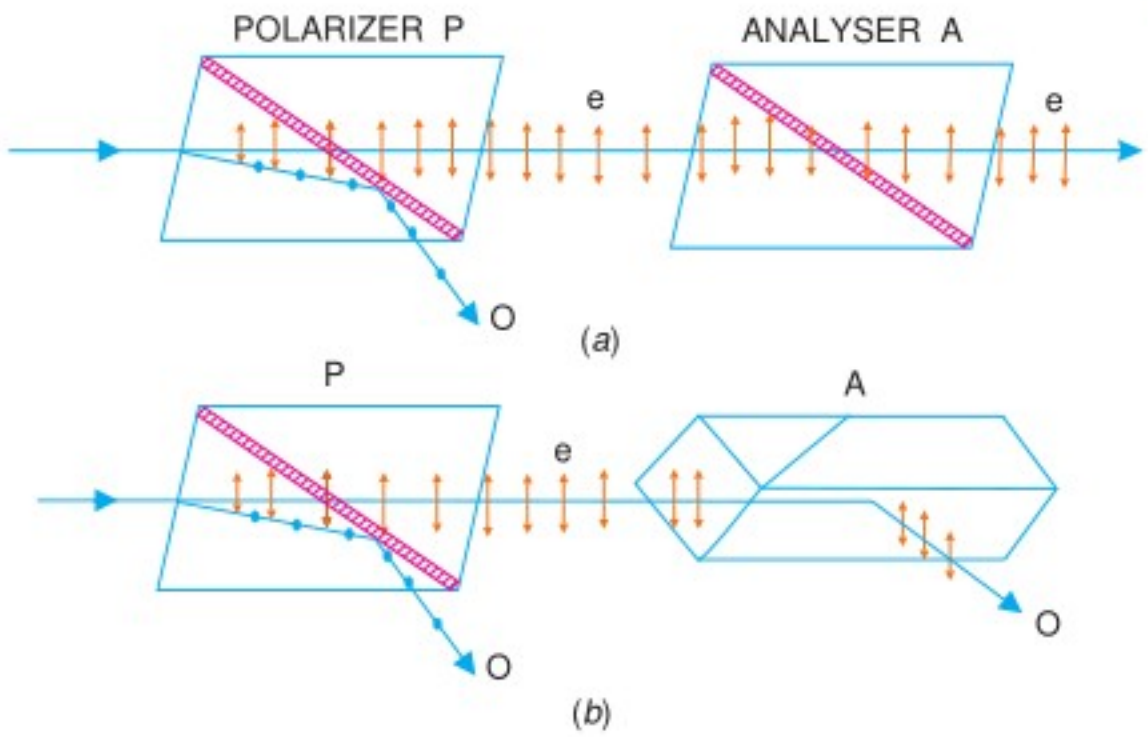


Fig. 20.30

P (Fig.20.30 b), the vibrations of the ray, emerging from P and incident on A, will be perpendicular to the principal section of A. In this position the ray behaves as o-ray inside the prism A and is totally internally reflected by the Canada balsam layer. Hence no light is transmitted by the prism A. In this configuration, the two Nicol prisms P and A are said to be **crossed.** If the Nicol prism A is further rotated through another 90°, the intensity of light emerging from A will go on increasing. The intensity will become a maximum when its principal section is again parallel to that of the prism P. Thus, the prism P produces linearly polarised light while the prism A detects it. Hence the prism P is called a polarizer and the prism A an analyser.

20.13. EFFECT OF POLARIZER ON LIGHT OF DIFFERENT POLARIZATIONS

The action of a polarizer, whether a Nicol prism or a Polaroid sheet, on light of different types of polarization is as follows:

(*i*) If **unpolarized light** is incident on a polarizer, it transmits half the intensity of light incident on it. The transmitted light stays constant in intensity on rotation of the polarizer.

(*ii*) If **partially polarized light** is incident on the polarizer, the intensity of the transmitted light depends on the direction of the transmission axis of the polarizer. The intensity varies from a maximum value I_{max} to a minimum value I_{min}. Two positions of I_{max} and two positions of I_{min} occur in one full rotation of the polarizer.

(*iii*) If **plane polarized light** is incident on the polarizer, the intensity of the transmitted light varies from zero to a maximum twice in one full rotation of the polarizer.

(*iv*) When **circularly polarized light** is incident on the polarizer, the transmitted light remains constant in intensity on rotation of the polarizer. This is interpreted as follows: the circular vibrations may be resolved into two mutually perpendicular linear vibrations of equal amplitude. When the circularly polarized light is incident on the polarizer, the vibration parallel to its transmission axis passes through the polarizer while the perpendicular component is not allowed. This happens in all positions of the polarizer in its rotation. Therefore, the intensity of the transmitted light stays constant in one full rotation of the polarizer.

(*v*) When **elliptically polarized light** is incident on the polarizer, the transmitted light varies in intensity from a maximum value I_{max} to a minimum value I_{min} on rotation of the polarizer (Fig.20.31). occurs when the transmission axis of the polarizer coincides with the semi-major axis of the ellipse and occurs when the transmission axis coincides with the semi-minor axis of the ellipse.

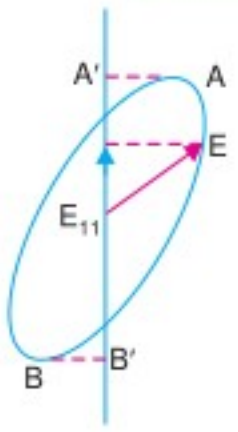


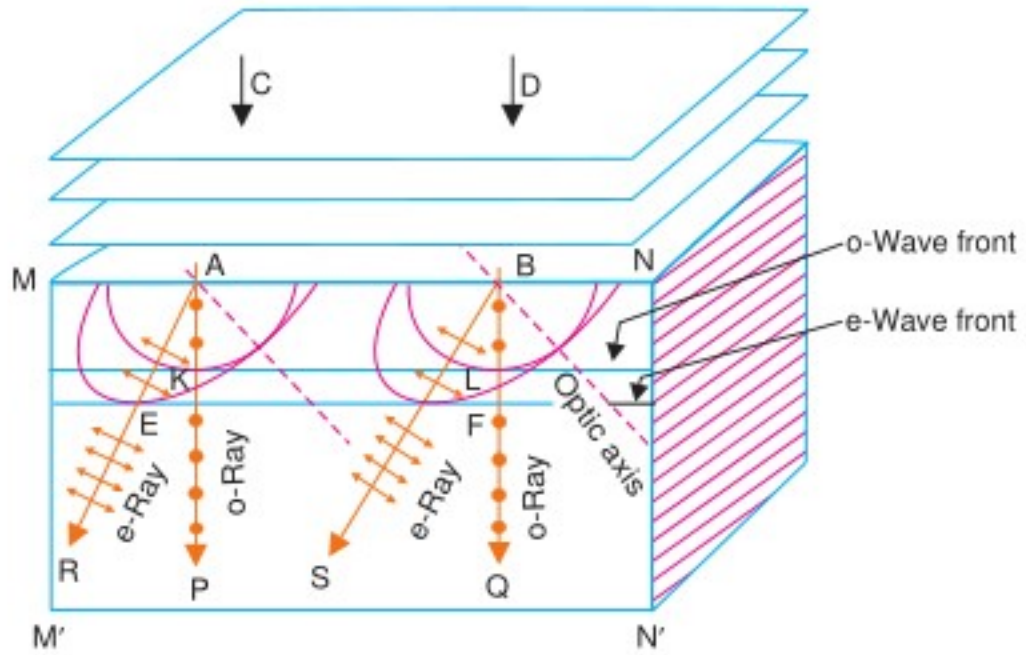
Fig. 20.31

20.14. HUYGENS' CONSTRUCTION OF WAVEFRONTS:

A thinner rectangular cross-section of a crystal can be cut from a bigger crystal in three different ways. It can be cut in such a way that the optic axis lies (*i*) inclined to the refracting face or (*ii*) parallel to the refracting face or (*iii*) perpendicular to the refracting face. The path of o-ray and e-ray within a uniaxial crystal can be determined using Huygens' principle of secondary wavelets. We take the example of a negative crystal for the purpose of tracing the paths of o-ray and e-ray.

Case.1. Optic axis inclined to the refracting edge:

Let MNN′M′ represent the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***inclined to the refracting edge.*** According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.



Unpolarized light beam is incident at normal on a calcite crystal slice. Optic axis is inclined to the incident ray. Huygens wavelets showing double refraction, and difference in velocities of *o*-ray and *e*-ray.

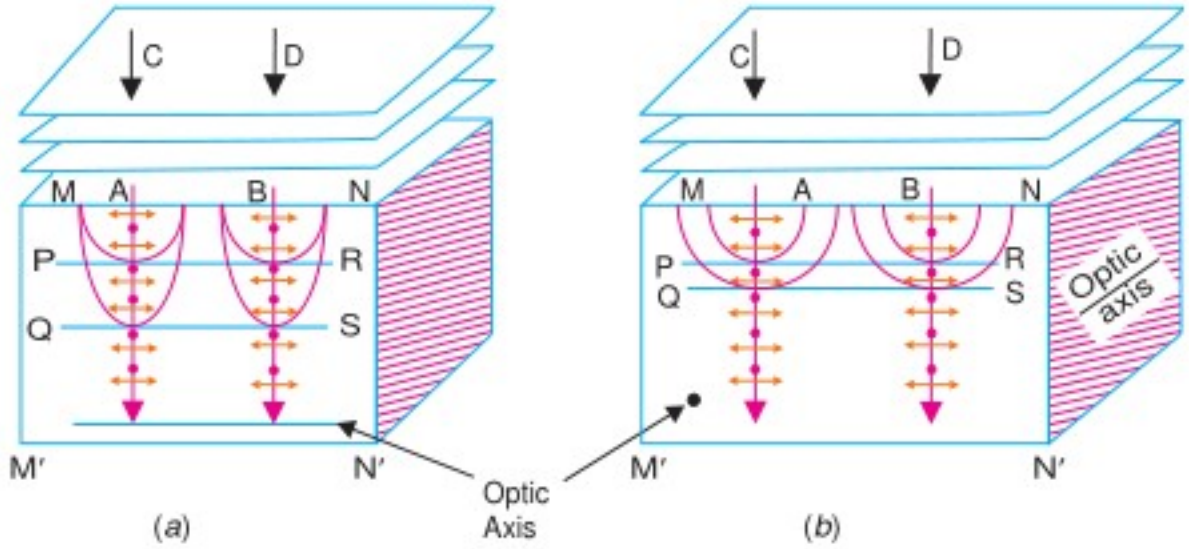
Fig. 20.32

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wave front becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration.

According to Huygens principle the points A and B produce elliptical and spherical wavelets. The position of the two wave fronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly, a circle of the same radius is drawn with B as centre. The ellipsoidal wave front can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. The circle and the ellipse touch each other, as required, along the optic axis. If we now draw the common tangents to the secondary wavelets, they represent the plane wave fronts corresponding to the two rays. Thus KL is the tangent to the spherical wavelets and EF is the tangent to the ellipsoidal wave fronts. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to K and B to L, we obtain the direction of propagation of o-ray. Similarly, the lines AP and BQ show the direction of propagation of e-ray. We find that in this case the o- ray and e-ray travel along different directions with different velocities.

Case 2(*a*): Optic axis in the plane of incidence and parallel to the refracting edge:

Let MNN′M′ represent the refracting face of the calcite (negative) crystal. Let CD represent the monochromatic plane wavefront incident normal to the crystal surface. Let the optic axis be ***parallel*** to the refracting edge MN and lie in the plane of incidence. See Fig.20.33 (a). According to Huygens the phenomenon of double refraction involves propagation of light waves of two types. The wave known as o-wave has a spherical wavefront whereas the e-wave has an ellipsoidal wavefront. The two wavefronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.



Unpolarized light beam incident normally on a calcite slab at right angles to the optic axis. Double refraction does not occur but *o*-ray and *e*-ray travel with different velocities in the crystal. (*a*) Optic axis in the plane of incidence and parallel to the crystal surface; (*b*) Optic axis perpendicular to the plane of incidence and parallel to the crystal surface.

Fig. 20.33

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wavefront becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration. According to Huygens' principle the points A and B produce elliptical and spherical wavelets. The two wavelets touch each other along AM and BN, which is the direction of optic axis, as required. The wave fronts at any instant can be drawn as tangential surfaces to the two wavelets. Note that the two wave fronts here are parallel to each other. The position of the two wavefronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly,

a circle of the same radius is drawn with B as centre. The ellipsoidal wavefront can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. PR is the tangent to the spherical wavelet and therefore represents the refracted wavefront for the o-ray. Similarly, QS represents the refracted wavefront for e-ray. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to P and B to R, we obtain the direction of propagation of o-ray. By joining A to Q and B to S we get the direction of propagation of e-ray. We find that in this case the o- ray and e-ray travel along the same direction but have different velocities.

Case 2 (*b*): Optic axis perpendicular to the plane of incidence and parallel to the refracting edge:

Let MN be the refracting edge of the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***parallel to the refracting edge but is perpendicular to the plane of incidence*** (Fig.20.33 b). According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.

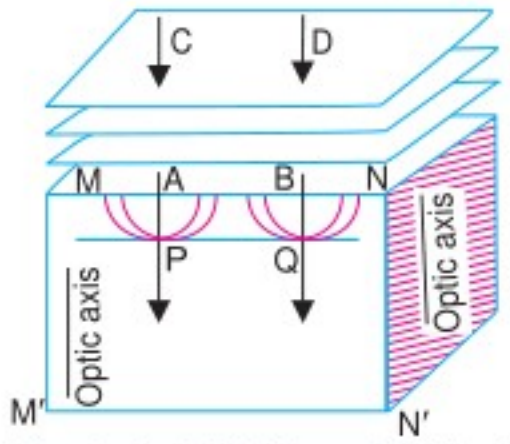
Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wave front becomes a source of secondary disturbance. The points A and B are chosen for the purpose of illustration. According to Huygens principle the points A and B produce elliptical and spherical wavelets. Since the spherical and ellipsoidal wavelets touch each other along a line perpendicular to the plane of incidence, the wavelets appear spherical in the plane of incidence. The position of the two wave fronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly, a circle of the same radius is drawn with B as centre. The ellipsoidal wave front can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. The circle and the ellipse touch each other, as required, along the optic axis, which is perpendicular to the surface MN. If we now draw the common tangents to the secondary wavelets, they represent the plane wavefronts corresponding to the two rays. Thus PR is the common tangent to the spherical wavelets and therefore represents the refracted wave front for the o-ray. Similarly, QS represents the refracted wave front for e-ray. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to P and B to R, we obtain the direction of propagation of o-ray. By joining A to Q and B to S we get the direction of propagation of e-ray. We find that in this case the o- ray and e-ray travel along the same direction but have different velocities.

Case 3 : Optic axis perpendicular to the refracting edge and lying in the plane of incidence:

Let MNN′M′ represent the calcite (negative) crystal. Let CD represent the monochromatic plane wave front incident normal to the crystal surface. The optic axis is ***perpendicular to the refracting edge and lies in the plane of incidence***. According to Huygens the phenomenon of double refraction involves two types of propagation of light waves. The wave known as o-wave has a spherical wave front whereas the e-wave has an ellipsoidal wave front. The two wave fronts touch each other along the optic axis of the crystal. In a negative crystal the e-ray velocity is greater than the o-ray velocity and hence the ellipsoidal surface lies outside the spherical surface.

Let us consider that parallel beam of light CD falls **normally** on the surface of the negative crystal. As soon as the parallel beam strikes the crystal boundary, each point on the wavefront becomes a new point source of light. The points A and B are chosen for the purpose of illustration. According

to Huygens' principle the points A and B produce elliptical and spherical wavelets. Since the two wavelets must touch each other along the optic axis, they touch at points P and Q (Fig.20.34). Note that the section of the ellipsoid lies outside the section of the circle. The minor axis of the ellipse and the radius of the circle are equal. The position of the two wave fronts after a lapse of 't' seconds can be determined as follows. A circle of radius '$\upsilon_o t$' is drawn taking A as centre. Similarly, a circle of the same radius is drawn with B as centre. The ellipsoidal wave front can be drawn if the major and minor axes are known. In a negative crystal the major axis is equal to '$2\upsilon_e t$' and minor axis is '$2\upsilon_o t$'. The circle and the ellipse touch each other, as required, along the optic axis, which is perpendicular to the refracting edge MN. If we now draw the common tangents to the secondary wavelets, they represent the plane wavefronts corresponding to the two rays. Thus PQ is the common tangent to the spherical wavelets as well as the ellipsoidal wave fronts. If now we join the point of origin of wavelets to the points of tangency, the direction of propagation of the rays will be known. Thus, if we join A to P and B to Q, we obtain the direction of propagation of o-ray as well as e-ray. We find that in this case the o- ray and e-ray travel along the same direction with the same velocity.



Unpolarized light beam incident normally on a calcite slab parallel to the optic axis. The o-ray and e-ray travel along the same velocity.

Fig. 20.34

20.11. EXPERIMENTAL DETERMINATION OF PRINCIPAL REFRACTIVE INDICES

For determining the refractive index for the extraordinary ray a calcite crystal is cut in the form of a prism with the optic axis perpendicular to the refracting edge of the prism and perpendicular to the base BC (Fig.20.35). It can also be cut with the optic axis parallel to the refracting edge of the prism. The prism is placed on the spectrometer table and is adjusted for the minimum deviation position for the extraordinary rays. The angle of minimum deviation δ_m is determined and the principal refractive index for the extraordinary ray is calculated from the relation (20.16a).

$$\mu_e = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}} \qquad \ldots (20.16\ a)$$

For a given wavelength, the ordinary and the extraordinary rays are separated while passing through the prism. Therefore, the angle of minimum deviation δ_m' for the ordinary ray can be measured and thus its refractive index can be calculated from the relation (20.16b).

$$\mu_0 = \frac{\sin\left(\frac{A+\delta'_m}{2}\right)}{\sin\frac{A}{2}} \qquad \ldots (20.16\ b)$$

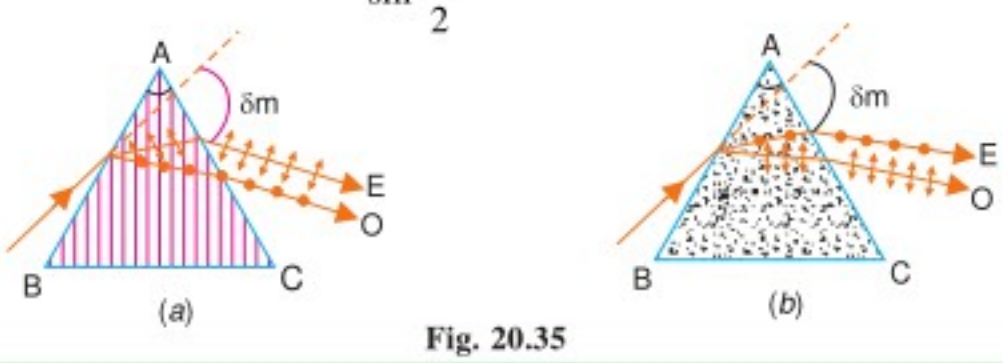
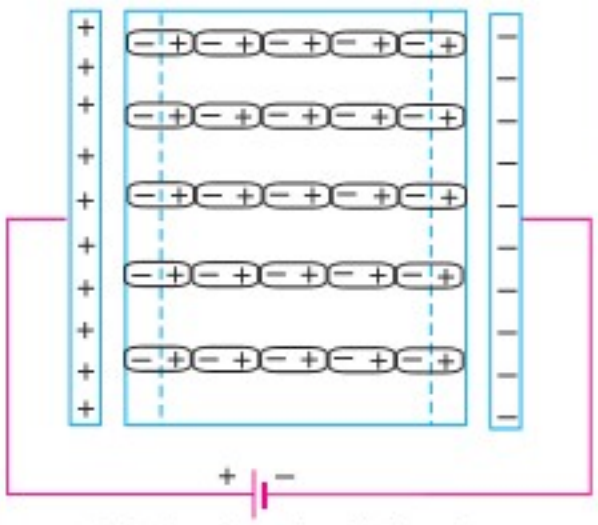


Fig. 20.35

20.16. ELECTROMAGNETIC THEORY OF DOUBLE REFRACTION:

The birefringent crystals belong to the group of non-conducting materials. They are known as dielectric materials. If a dielectric crystal is placed between the plates of a parallel plate capacitor and a voltage is applied, the crystal becomes polarized. This polarization is dielectric polarization and is different from the optical polarization. The voltage produces an electric field E and under the action of the electric field, electron clouds in the atoms of the crystal distort with the result that the centres of action of negative charge and positive charge get separated by a small distance. It means that electrical dipoles are produced throughout the crystal. Negative charges of the dipoles lie on the side of the positively charged plate of the capacitor and positive charges lie on the side of the negatively charged plate. Thus electric charges of opposite nature are induced on the opposite faces of the crystal in response to the applied electric field (see Fig.20.36). This is known as **dielectric polarization.**



Polarization of a dielectric

Fig. 20.36

The electric field, polarization and displacement in the crystal are related by

$$D = \varepsilon_o E + P \qquad \ldots (20.17)$$

where

$$D = \varepsilon_o \varepsilon_r E \qquad \ldots (20.18)$$

and

$$P = \chi \varepsilon_0 E \qquad \ldots (20.19)$$

ε_r and χ are known as *relative permittivity* and *electric susceptibility* of the crystal. The relative permittivity is related to the refractive index of the material through the relation

$$\mu = \sqrt{\varepsilon_r} = \sqrt{\varepsilon / \varepsilon_o} \qquad \ldots (20.20)$$

We know that refractive index is related to the velocity of light in the medium.

$$\mu = \frac{c}{\upsilon} \qquad \ldots (20.21)$$

It is not necessary for dielectric polarization that the crystal be placed in between the plates of a capacitor. The electric field of a light wave propagating in the crystal also can cause dielectric polarization. According to the above relations the dielectric polarization caused by the light wave is related to the velocity (direction of propagation) of light wave with in the crystal.

In case of isotropic crystals, the vectors *D, E* and *P* are parallel and ε_r, μ and χ are scalars. In anisotropic crystals the induced polarization is in a direction different from that of the field. As a result, ε_r, μ and χ are tensor quantities. The components of polarization are given by

$$\begin{aligned} P_x &= \varepsilon_o \left(\chi_{11} E_x + \chi_{12} E_y + \chi_{13} E_z\right) \\ P_y &= \varepsilon_o \left(\chi_{21} E_x + \chi_{22} E_y + \chi_{23} E_z\right) \\ P_z &= \varepsilon_o \left(\chi_{31} E_x + \chi_{32} E_y + \chi_{33} E_z\right) \end{aligned} \qquad (20.22)$$

The above equations can be simplified by choosing the coordinate axes in such a way that the off-diagonal elements vanish. Then

$$\begin{aligned} P_x &= \varepsilon_o \chi_{11} E_x \\ P_y &= \varepsilon_o \chi_{22} E_y \end{aligned} \qquad (20.23)$$

$$P_z = \varepsilon_o \chi_{33} E_z$$

Similarly, we can express $D_x = \varepsilon_{11} E_x$,

$$D_y = \varepsilon_{22} E_y \qquad \ldots (20.24)$$

$$D_z = \varepsilon_{33} E_z$$

These directions are called the **principal axes** of the crystal and the corresponding diagonal terms $\varepsilon_{11}, \varepsilon_{22}$ and ε_{33} the *principal permittivities*. The variation in the permittivity and the corresponding variation in refractive index and hence in wave velocity is the origin of double refraction. μ_x, μ_y and μ_z are the three *principal refractive indices* corresponding to the three principal permittivities. **The subscripts *x*, *y* and *z* relate to the direction of polarization of the light waves and not to their direction of propagation.** The principal permittivities and refractive indices are related as

$$\mu_x = \sqrt{\varepsilon_{11}/\varepsilon_o}\,, \quad \mu_y = \sqrt{\varepsilon_{22}/\varepsilon_o} \quad \text{and} \quad \mu_z = \sqrt{\varepsilon_{33}/\varepsilon_o} \qquad \ldots (20.25)$$

20.16.1 INDEX ELLIPSOID OR OPTICAL INDICATRIX:

The different speeds of light in different directions in a crystal are conveniently represented by a three-dimensional figure that shows the refractive index for light waves in their direction of polarization. The representation is known as the **indicatrix**. *The optical indicatrix is spherical or ellipsoidal surface having the three different refractive indices as axes.*

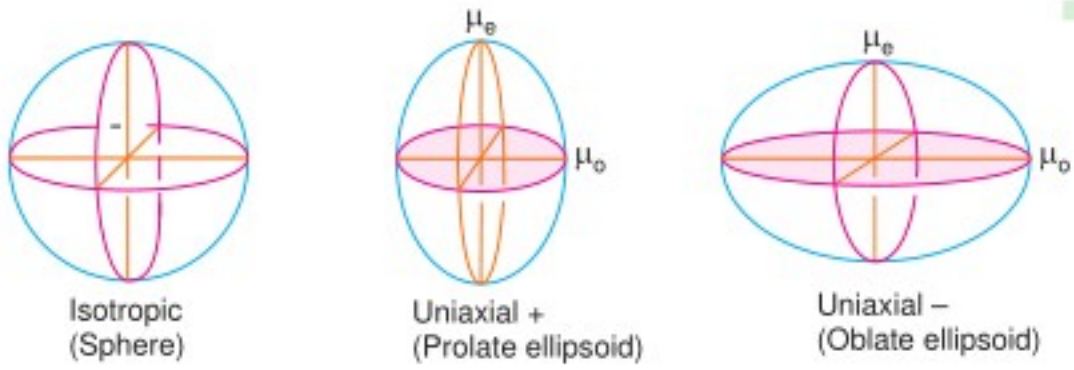


Fig. 20.37

In case of isotropic crystals, the axes are all of the same length and consequently the refractive indices in three directions are the same. Therefore, the indicatrix is a *sphere*, as shown in Fig. 20.37 (a). In case of uniaxial crystals, a unique axis exists normal to the plane of two equal axes. The velocity of propagation of light along this unique axis (z-direction) is independent of the wave polarisation and this direction is the *optic axis* direction. The refractive index along this direction is μ_e. In a plane perpendicular to the optic axis, the refractive index is independent of direction, that is $\mu_x = \mu_y$ denoted by μ_o. Therefore, in uniaxial crystals the indicatrix is an **ellipsoid** having a circular cross-section in the *xy*-plane. The ellipsoid is prolate if $\mu_e > \mu_o$ and oblate if $\mu_e < \mu_o$ (see Fig. 20.37b).

The form of optical indicatrix can be obtained as follows. The energy density in a dielectric medium is given by

$$W = \tfrac{1}{2} D \cdot E \qquad \ldots (20.26)$$

$$W = \tfrac{1}{2}\left[\frac{D_x^2}{\varepsilon_{11}} + \frac{D_y^2}{\varepsilon_{22}} + \frac{D_z^2}{\varepsilon_{33}}\right] \qquad \ldots (20.27)$$

or $$\frac{1}{2}\left[\frac{D_x^2}{W\varepsilon_{11}}+\frac{D_y^2}{W\varepsilon_{22}}+\frac{D_z^2}{W\varepsilon_{33}}\right]=1$$

or $$\left[\frac{D_x^2}{2W\mu_x^2\varepsilon_o}+\frac{D_y^2}{2W\mu_y^2\varepsilon_o}+\frac{D_z^2}{2W\mu_z^2\varepsilon_o}\right]=1$$

Writing $x^2 = D_x^2/2\varepsilon_o W$, $y^2 = D_y^2/2\varepsilon_o W$ and $z^2 = D_z^2/2\varepsilon_o W$, we get

$$\frac{x^2}{\mu_x^2}+\frac{y^2}{\mu_y^2}+\frac{z^2}{\mu_z^2}=1 \qquad \ldots (20.28)$$

This is the equation of an ellipsoid with semi-axes μ_x, μ_y and μ_z. In the case of a uniaxial crystal the indicatrix has circular symmetry about the z-axis. In this case,

$$\frac{x^2}{\mu_0^2}+\frac{y^2}{\mu_0^2}+\frac{z^2}{\mu_e^2}=1 \qquad \ldots (20.29)$$

Let us now consider light propagating in a direction **r**, at an angle θ to the optic axis. Because of the circular symmetry, we can choose that the y-axis should coincide with the projection of **r** on the *xy*-plane. The plane normal to *r* intersects the ellipsoid in the shaded ellipse. The two allowed directions of polarization are parallel to the axes of the ellipse and thus correspond to *OP* and *OQ* (Fig.20.38). They are thus perpendicular to *r* as well as to each other. The two waves polarized along these directions have refractive indices given by $OP = \mu_0$ and $OQ = \mu_e(\theta)$. In case of the e-ray, the plane of polarization varies with θ as does the refractive index.

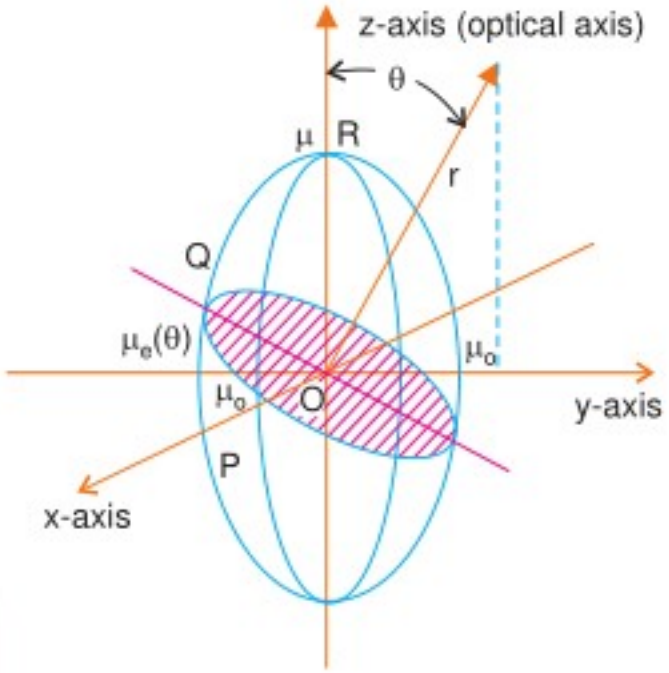


Fig. 20.38

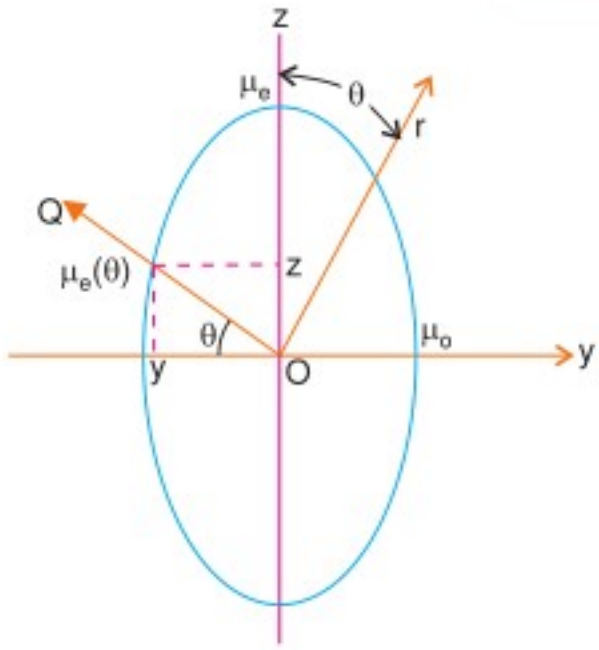


Fig. 20.39

We can determine the relationship between $\mu_e(\theta)$, μ_e and μ_o with the help of the Fig.20.39. The figure shows the intersection of the indicatrix with the yz-plane. From the diagram, it is seen that

$$\mu_e^2(\theta) = z^2 + y^2 \qquad \ldots (20.30)$$

and $$z = \mu_e(\theta)\sin\theta \qquad \ldots (20.31)$$

$$\therefore \quad \mu_e^2(\theta) = \mu_e^2(\theta)\sin^2\theta + y^2$$

or $$y^2 = \mu_e^2(\theta)\cos^2\theta \qquad \ldots (20.32)$$

The equation of the ellipse shown in Fig. 20.39 is

$$\frac{y^2}{\mu_o^2}+\frac{z^2}{\mu_e^2}=1 \qquad (20.23)$$

Substituting equ. (20.31) and equ. (20.32) into equ. (20.33), we obtain

$$\frac{1}{\mu_e^2(\theta)}=\frac{\cos^2\theta}{\mu_o^2}+\frac{\sin^2\theta}{\mu_e^2} \qquad (20.34)$$

Thus, for $\theta = 0^o$, that is propagation along the optic axis, $\mu_e(0°) = n_0$ while for $\theta = 90^o$, $\mu_e(90°) = \mu_e$. The two polarizations which can be propagated correspond to the maximum and minimum refractive indices given by the index ellipsoid. For propagation parallel to the optic axis (that is z-direction), there is no birefringence as the section of the ellipsoid perpendicular to this direction is a circle. For propagation perpendicular to the optic axis, the birefringence will be a maximum, the permitted polarizations will be parallel to the y-axis with refractive index μ_o and parallel to the z-axis with refractive index μ_e.

20.17. PHASE DIFFERENCE BETWEEN e-RAY AND o-RAY

We have seen that natural light incident on the surface of an anisotropic crystal undergoes double refraction and produces two plane polarized waves (Fig.20.40a). Let us consider the particular case of a slice of a calcite crystal (a negative uniaxial crystal) where the optic axis is parallel to refracting face of the crystal. The two waves travel along the same direction in the crystal but with different velocities. As a result, when the waves emerge from the rear face of the crystal, an optical path difference would have developed between them (Fig.20.40b). The optical path difference can be calculated as follows:

Let d be the thickness of the crystal.

The optical path for o-ray within the crystal $= \mu_o d$

The optical path for e-ray within the crystal $= \mu_e d$

$\therefore$ The optical path difference between e-ray and o -ray $\Delta = (\mu_o - \mu_e)\, d$ ... (20.35)

Consequently, a phase difference arises between the two waves. It is given by

$$\delta = \frac{2\pi}{\lambda}(\mu_o - \mu_e)\, d \qquad \text{... (20.36)}$$

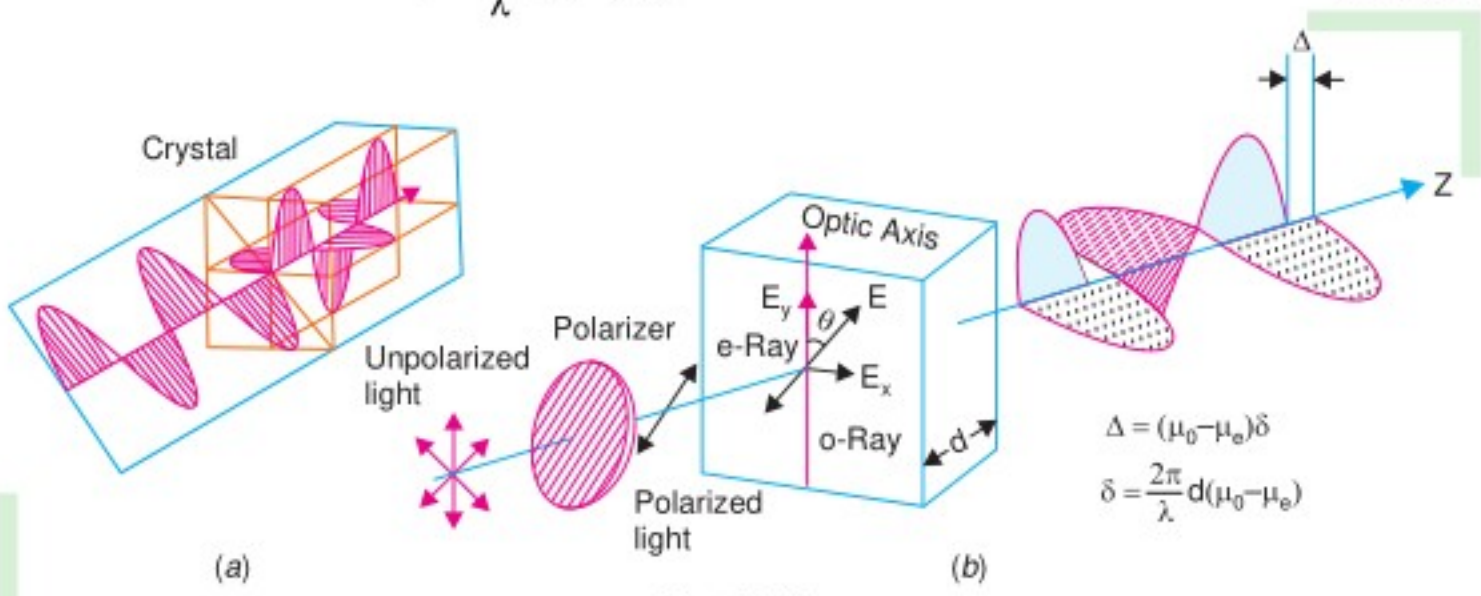


Fig. 20.40

As the two component waves are derived from the same incident wave, the two waves are in phase at the front face and have emerged from the crystal with a constant phase difference and hence it may be expected that the waves are in a position to interfere with each other. However, as the planes of polarization of o-ray and e-ray are perpendicular to each other, interference cannot take place between e- and o-rays. The waves instead combine with each other to give elliptically polarized wave.

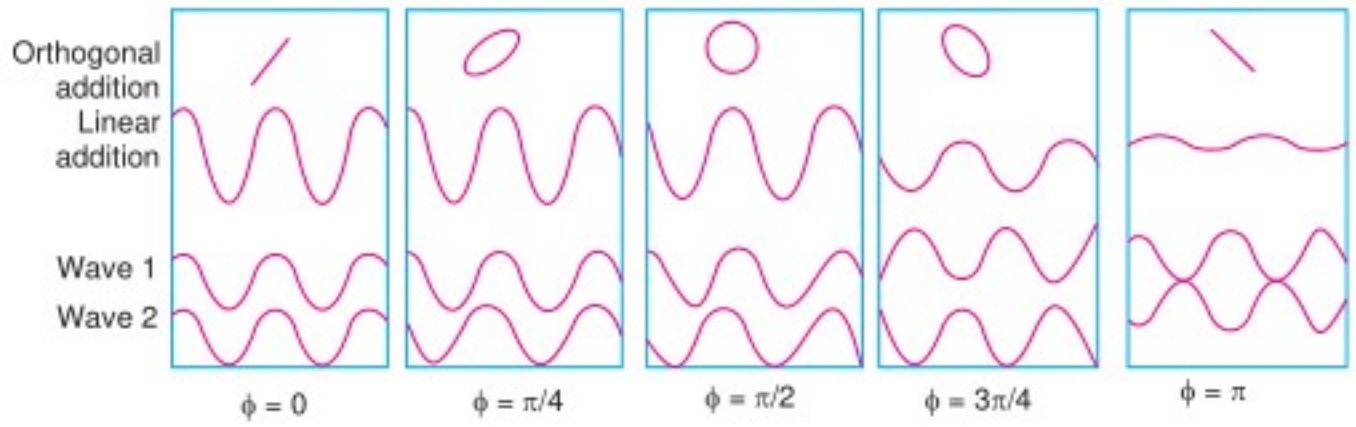


Fig. 20.41

The most important point to be noted here is as follows: The superposition of two coherent waves having a *common plane* of polarization yields a wave polarized in the same plane. The superposition leads to linear addition of waves, i.e., **interference**. The resultant vibration is *linear* and occurs in the same direction as that of the two superposing vibrations (see Fig.20.41). On the other hand, if the two coherent waves are polarized in orthogonal planes, the resultant vibrational motion takes place in two dimensions either in the form of an ellipse, a circle or a straight line depending on the phase difference between the waves (Fig.20.41). This is known as *orthogonal addition.*

20.18. SUPERPOSITION OF WAVES LINEARLY POLARISED AT RIGHT ANGLES

Let us now look at the result of superposition of two waves linearly polarized at right angles to each other (see Fig.20.42 a). Let us consider two light waves travelling in the same direction, x ; one wave is polarized in the xy-plane and the other is polarized in yz-plane. We are interested to know the state of polarization of the resultant wave at the plane x = *constant*.

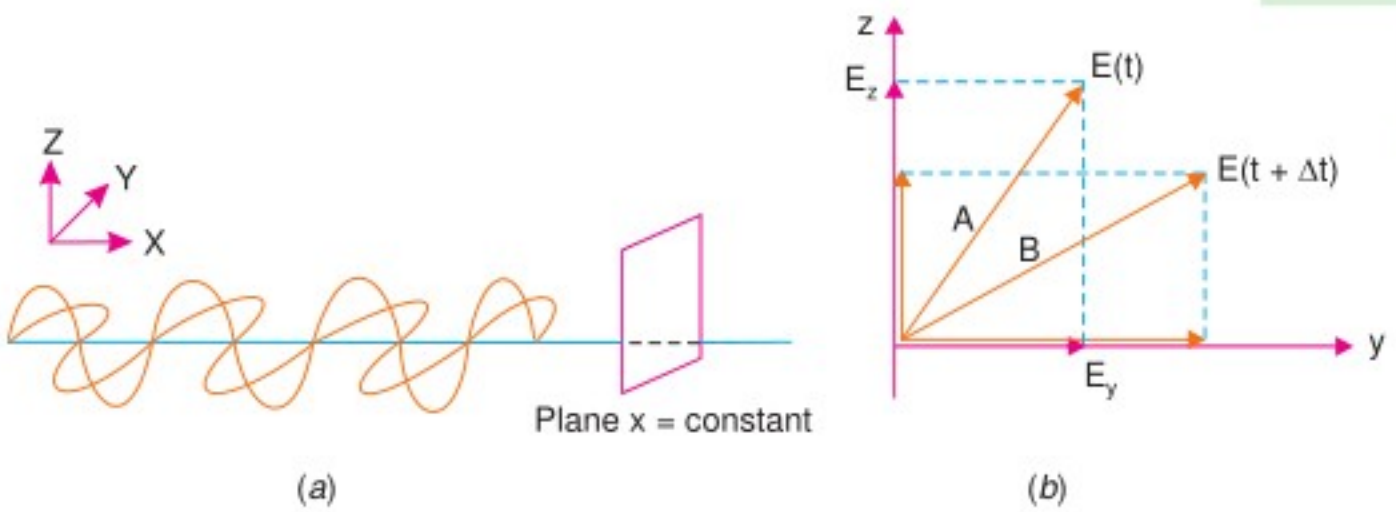


Fig. 20.42

Let the two orthogonal waves be represented by

$$E_y = E_1 \cos\left(k\,x - \omega\,t\right) \tag{20.37}$$

$$E_z = E_2 \cos\left(k x - \omega t + \delta\right) \qquad \ldots (20.38)$$

The waves are of the same frequency $\nu = \omega/2\pi$. δ is the phase difference between the waves. At a given time t, the optical vectors E_y and E_z produce a resultant optical vector of magnitude, say A and at a slightly later time, $t + \Delta t$, they give rise to a resultant vector of magnitude B which points in a different direction (see Fig.20.42b). With progress of time the tip of the resultant optical vector moves along a curve in the yz-plane. We apply the *principle of superposition* to find the equation of the curve traced by the tip of the resultant vector of the two vectors.

According to the principle of superposition

$$\begin{aligned} E &= E_y + E_z \\ &= E_1 \cos(kx - \omega t) + E_2 \cos(kx - \omega t + \delta) \end{aligned} \qquad \ldots (20.39)$$

The equation of the curve may be found by eliminating 't' from the equations. We can write the expansion of equ. (20.39) as

$$\begin{aligned} E_z &= E_2 \cos(kx - \omega t)\cos\delta - E_2 \sin(kx - \omega t)\sin\delta \\ &= E_2 \cos(kx - \omega t)\cos\delta \pm [1 - \cos^2(kx - \omega t)]^{1/2} E_2 \sin\delta \end{aligned} \qquad \ldots (20.40)$$

We find from equ.(20.37) that

$$\cos(kx - \omega t) = E_y/E_1$$

$$\therefore \quad E_z = E_2 \frac{E_y}{E_1} \cos\delta \pm \sqrt{1 - \frac{E_y^2}{E_1^2}}\; E_2 \sin\delta \qquad \ldots (20.41)$$

Rearranging the terms, we get

$$\left[E_z - \frac{E_2}{E_1} E_y \cos\delta\right] = \pm\sqrt{1 - \left(\frac{E_y}{E_1}\right)^2}\; E_2 \sin\delta$$

On squaring both the sides, we obtain

$$E_z^2 + \frac{E_y^2 E_2^2}{E_1^2}\cos^2\delta - \frac{2E_y E_z E_2}{E_1}\cos\delta = E_2^2 \sin^2\delta - \frac{E_y^2 E_2^2}{E_1^2}\sin^2\delta$$

Rearranging the terms, we get

$$E_z^2 + \frac{E_y^2 E_2^2}{E_1^2}(\cos^2\delta + \sin^2\delta) - \frac{2E_y E_z E_2}{E_1}\cos\delta = E_2^2 \sin^2\delta \qquad \ldots (20.42)$$

$$E_z^2 + \frac{E_y^2 E_2^2}{E_1^2} - \frac{2E_y E_z E_2}{E_1}\cos\delta = E_2^2 \sin^2\delta$$

Dividing both the sides by E^2_2 and rearranging the terms, we obtain

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} - \frac{2E_y E_z}{E_1 E_2}\cos\delta = \sin^2\delta \qquad \ldots (20.43)$$

Equation (20.43) is the general equation of an *ellipse*. Hence, the tip of the resultant vector traces an ellipse in the yz-plane. The ellipse is constrained within a rectangle having sides $2E_1$ and $2E_2$. The major axis makes an angle α with the y-axis (see Fig.20.43).

$$\tan 2\alpha = 2E_1 E_2 \cos\delta / \left(E_1^2 - E_2^2\right)$$

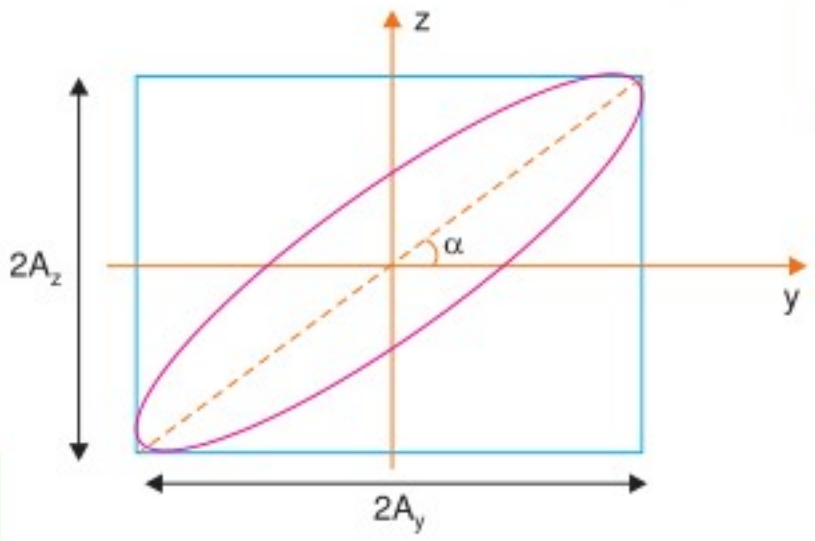


Fig. 20.43

Special cases:

1. When $\delta = 0$, or $\pm 2m\pi$, the two waves are in phase.
   $\cos\delta = 1$ and $\sin\delta = 0$ and the equ. (20.43) reduces to

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} - \frac{2E_yE_z}{E_1E_2} = 0$$

$$\left[\frac{E_y}{E_1} - \frac{E_z}{E_2}\right]^2 = 0$$

$$\frac{E_y}{E_1} - \frac{E_z}{E_2} = 0$$

$$\therefore \quad E_z = \frac{E_2}{E_1}E_y \qquad \text{... (20.44)}$$

The above equation represents a straight line, having a slope (E_2/E_1). Therefore, the equation represents a wave having its plane of polarisation making an angle $\tan^{-1}(E_2 - E_1)$ with respect to the y-axis. It means that **the resultant of two plane-polarised waves, which are in phase (*i.e.*, coherent waves), is again a plane-polarised wave.**

2. When $\delta = \pi$, or $\pm (2m+1)\pi$, the two waves are in opposite phase.
   $\cos\delta = -1$ and $\sin\delta = 0$ and the equ. (20.43) reduces to

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} - \frac{2E_yE_z}{E_1E_2} = 0$$

$$\left[\frac{E_y}{E_1} + \frac{E_z}{E_2}\right]^2 = 0$$

$$\frac{E_y}{E_1} + \frac{E_z}{E_2} = 0$$

$$\therefore \quad E_z = -\frac{E_2}{E_1}E_y \qquad \text{... (20.45)}$$

This equation represents a straight line of a slope $(-E_2/E_1)$. Therefore, the equation represents a wave having its plane of polarization making an angle $\tan^{-1}(-E_2/E_1)$ with respect to the y-axis. It means that **the resultant of two plane-polarized waves, which are in opposite phase (*i.e.*, coherent waves), is again a plane-polarized wave.**

3. If $\delta = \pi/2$, or $\pm (2m+1)\pi/2$, then $\cos\delta = 0$ and $\sin\delta = 1$. Equ.(20.43) reduces to

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} = 1 \qquad \text{... (20.46)}$$

This is the equation of an ellipse whose major axis and minor axis coincide with y- and z-co-ordinate axes. Therefore, when the two plane polarized waves are out of phase by 90°, their resultant is an elliptically polarized wave.

4. In the particular case, when $\delta = \pi/2$ and $E_1 = E_2 = E_o$, equ.(20.43) reduces to

$$E_y^2 + E_z^2 = E_o^2 \qquad \text{... (20.47)}$$

This is the equation of a circle. Hence the resultant light is circularly polarized.

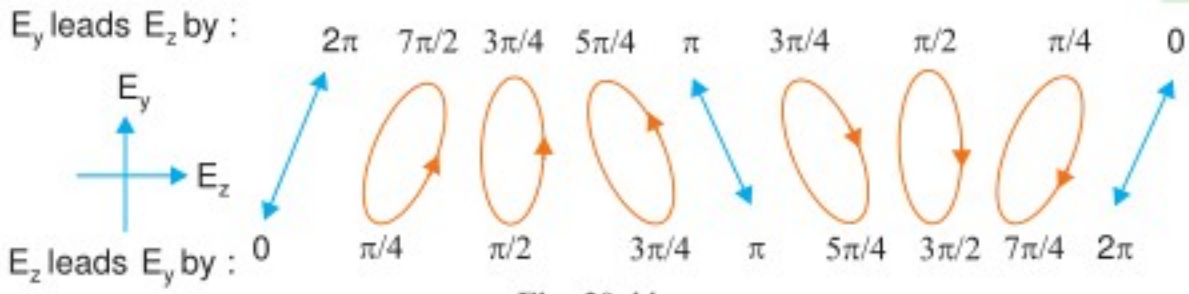


Fig. 20.44

Fig.20.44 shows more generally how the E vector changes with time for various values of δ.

20.18.1 SUPERPOSITION OF e-RAY AND O-RAY:

Now we are in a position to understand what happens when e-ray and o- ray overlap on each other after emerging from an anisotropic crystal plate. It is obvious that they cannot produce interference fringes as in a double slit experiment, since the o-ray are not vibrating in the same plane. On the other hand, they combine to produce different states of polarization depending upon their optical path difference.

1. When the optical path difference is 0 or an even or odd multiple of λ/2, the resultant light wave is linearly polarized.
2. When the optical path difference is λ/4, the resultant light wave is **elliptically polarized.**
3. In the particular instance when the wave amplitudes are equal and the optical path difference is λ/4, the resultant light wave is **circularly polarized**.

20.19. RETARDERS

A **retarder** is a uniform plate of birefringent material whose optic axis lies in the plane of the plate. Retarders are called quarter-wave plates, half-wave plates and full-wave plates depending on their action. They divide the incident wave into two polarized waves that travel perpendicular to the plate at different speeds. A phase retardation of one wave relative to the other is therefore introduced as the waves cross the thickness *d* of the plate. They are used to produce circularly or elliptically polarised light and to analyse polarised light into its elliptical components.

20.19.1 QUARTER WAVE PLATE

A *quarter wave plate* is a thin plate of birefringent crystal having the optic axis parallel to its refracting faces and its thickness adjusted such that it introduces a quarter-wave (λ/4) path difference (or a phase difference of 90°) between the e-ray and o-ray propagating through it.

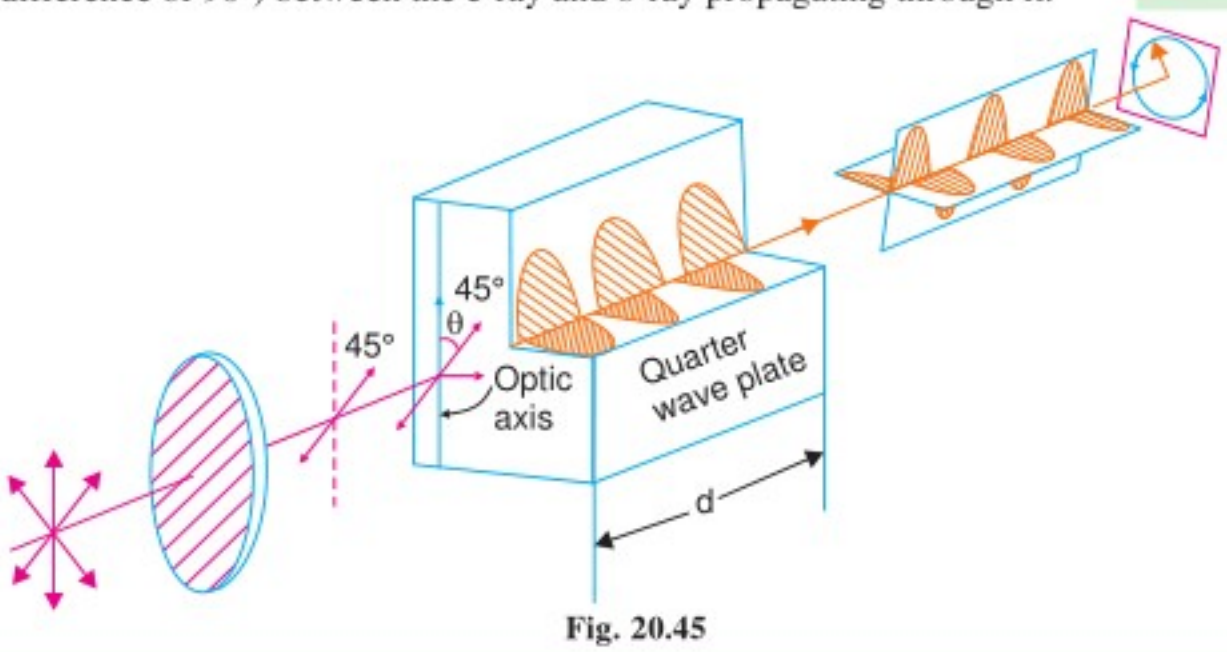


Fig. 20.45

When a plane polarized light wave is incident on a negative birefringent crystal having the optic axis parallel to its refracting face, the wave splits into e-wave and o-wave (Fig.20.45). The two waves travel along the same direction but with different velocities. As a result, when they emerge from the rear face of the crystal, an optical path difference would be developed between them. Thus,

$$\therefore \qquad (\mu_o - \mu_e)d = \frac{\lambda}{4} \qquad \ldots (20.48)$$

$$d = \frac{\lambda}{4(\mu_o - \mu_e)} \qquad \ldots (20.49)$$

A quarter wave plate introduces a phase difference δ, between e-ray and o-ray given by $\delta = (2\pi/\lambda)\Delta = \pi/2 = 90°$.

A quarter-wave plate is used for producing elliptically or circularly polarized light. It converts plane-polarized light into elliptically or circularly polarized light depending upon the angle that the incident light vector makes with the optic axis of the quarter wave plate.

20.19.2 HALF WAVE PLATE

A *half wave plate* is a thin plate of birefringent crystal having the optic axis parallel to its refracting faces and its thickness chosen such that it introduces a half-wave ($\lambda/2$) path difference (or a phase difference of 180°) between e-ray and o-ray.

When a plane polarized light wave is incident on a birefringent crystal having the optic axis parallel to its refracting faces, it splits into two waves: o- and e-waves. The two waves travel along the same direction inside the crystal but with different velocities. As a result, when they emerge from the rear face of the crystal, an optical path difference would be developed between them.

$$\therefore \qquad (\mu_o - \mu_e)d = \frac{\lambda}{2} \qquad \ldots (20.50)$$

$$\therefore \qquad d = \frac{\lambda}{2(\mu_o - \mu_e)} \qquad \ldots (20.51)$$

A half wave plate introduces a phase difference δ, between e-ray and o-ray given by $\delta = (2\pi/\lambda)\Delta = \pi = 180°$.

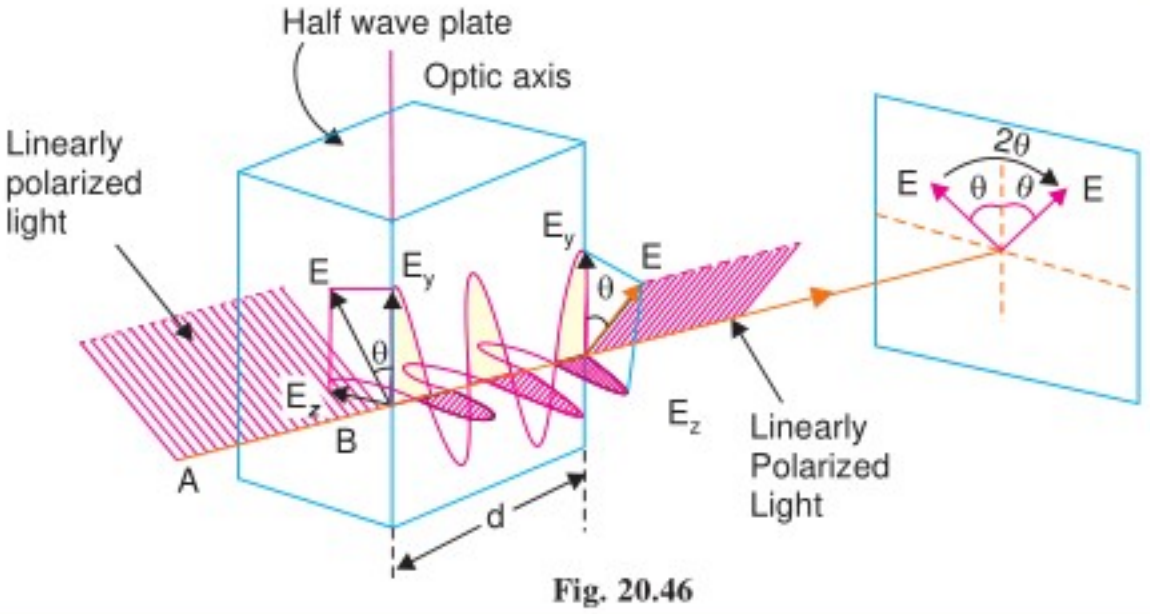


Fig. 20.46

Now let a plane polarized light be incident normally on the half-wave plate. Let the electric vector E make an angle with the optic axis of the half wave plate (See Fig.20.46). The incident wave splits into two waves, e- and o-waves. The waves progressively develop path difference as

they travel through the crystal and they emerge with a phase difference of 180°. When the two waves combine, they yield a plane-polarized wave, which has its plane of polarization rotated through an angle of 2θ.

Therefore, ***a half-wave plate rotates the plane of polarization of the incident plane polarized light through an angle 2θ.*** The half wave plate will invert the handedness of elliptical or circular polarized light, changing right to left and vice versa.

Now we are in a position to understand what happens when e-ray and o- ray overlap on each other after emerging from an anisotropic crystal plate. It is obvious that they cannot produce interference fringes as in a double slit experiment. On the other hand, they combine to produce different states of polarization depending upon their optical path difference.

1. When the optical path difference is 0 or an even or odd multiple of λ/2, the resultant light wave is linearly polarized.
2. When the optical path difference is λ/4, the resultant light wave is **elliptically polarized.**
3. In the particular instance when the wave amplitudes are equal and the optical path difference is λ/4, the resultant light wave is **circularly polarized**.

20.20. PRODUCTION OF ELLIPTICALLY POLARIZED LIGHT

A quarter wave plate and a polarizer are the optical devices necessary to produce elliptically polarized light from unpolarized light.

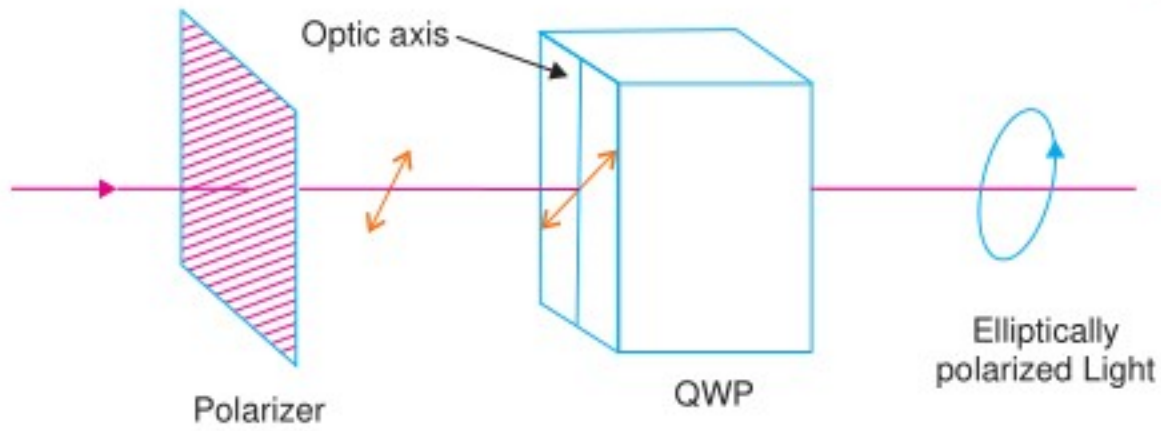


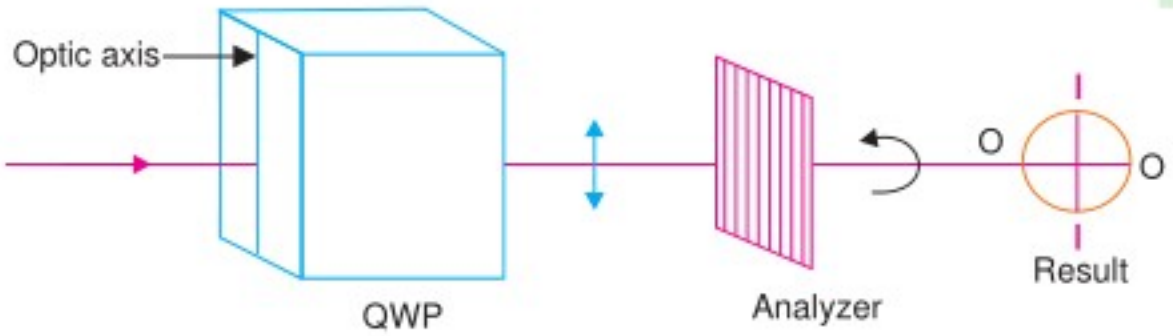
Fig. 20.47

Unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer (a polaroid sheet or a Nicol prism), as shown in Fig.20.47. The plane-polarized light is then made incident on a quarter wave plate. The quarter wave plate or the polarizer is rotated such that the electric vector E of plane polarized light wave makes an angle θ (≠45°) with the optic axis of the quarter wave plate. The incident ray divides into o-ray and e-ray of amplitudes E sin θ and E cos θ. The rays travel along the same direction in the crystal with different velocities. The two rays are polarized in orthogonal planes. They are in phase at the front face but progressively get out of phase as they travel through the crystal. When they emerge out of the crystal they will have a path difference of λ /4 or a phase difference of 90°. When they combine, they produce elliptically polarized light.

20.20.1 DETECTION OF ELLIPTICALLY POLARIZED LIGHT

The light beam is allowed to pass through an analyzer (a polaroid sheet or a Nicol prism). If on rotating the analyzing Polaroid sheet or Nicol, the intensity of the emerging beam varies from a maximum to a minimum value, but never reaching zero, then the incident light is elliptically polarized.

A similar result would be obtained if the incident light were partially polarized. The two cases may be distinguished by inserting a quarter wave plate in the path of light before it falls on the analyzer.



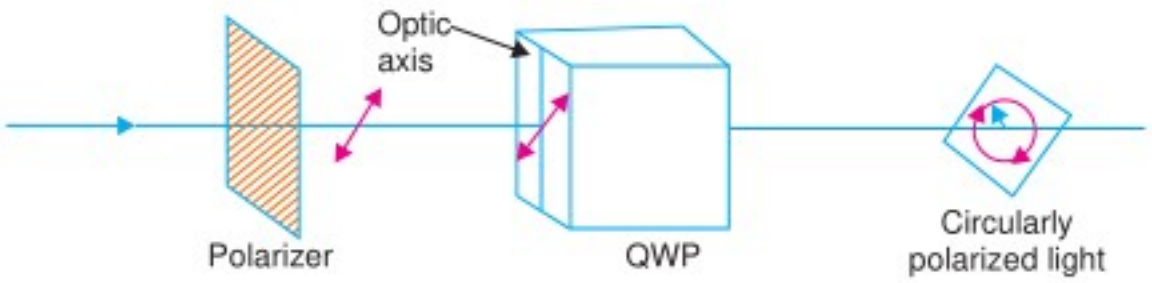
Detection of elliptically polarized light

Fig. 20.48

If the original light is elliptically polarized, it may be considered as resultant of two coherent plane polarized waves that is e-ray and o-ray, which are out of phase by 90°. If the light passes through the quarter wave plate, an additional phase difference of 90° is introduced between the e-ray and o-ray. Therefore, the total phase difference becomes 180° between the e-ray and o-ray. On emerging from the quarter plate, the e-and o-rays combine to produce plane-polarized light. If the light coming out of quarter wave plate is examined with an analyzer, light will be extinguished twice in one full rotation of the polarizer as shown in Fig.20.48.

20.21. PRODUCTION OF CIRCULARLY POLARIZED LIGHT

A quarter wave plate and a polarizer are the optical devices required for producing circularly polarized light from unpolarized light.



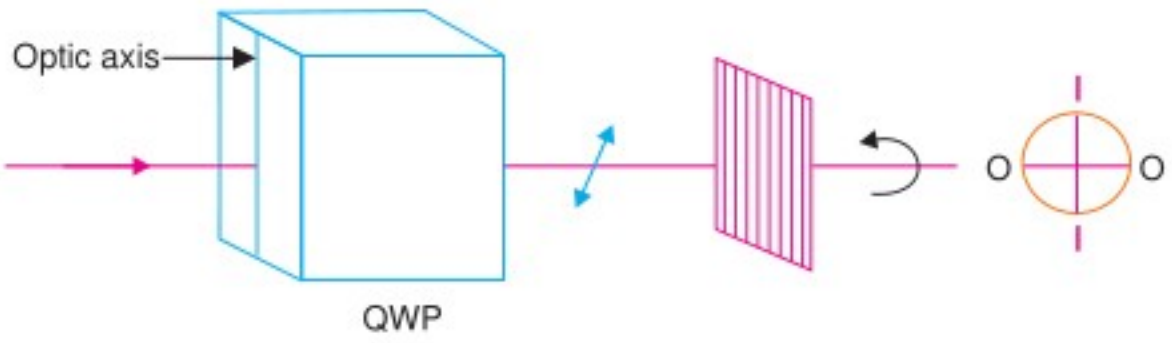
Production of Circularly Polarized Light

Fig. 20.49

Unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer (a polaroid sheet or a Nicol prism), as shown in Fig.20.49. Plane polarized light is then made incident on a quarter wave plate. The polarizer and the quarter wave plate are rotated such that the electric vector E of the plane-polarized wave makes an angle of 45° with the optic axis of the quarter wave plate. The plane polarized wave incident on the quarter wave plate splits into two rays, o-ray and e-ray of equal amplitude ($E_1 \cos 45° = E_2 \sin 45°$). The two rays travel in the same direction inside the crystal but with different velocities. The two rays are in phase at the front face of the crystal but progressively get out of phase as they travel through the crystal. As they emerge from the rear face of the crystal, they will have a path difference of $\lambda/4$ or phase difference of 90°. The two rays are linearly polarized in mutually perpendicular directions. When they combine, they produce circularly polarized light.

20.21.1 DETECTION OF CIRCULARLY POLARIZED LIGHT

The light beam is allowed to pass through an analyzer (a polaroid sheet or a Nicol prism). If on rotating the analyzing polaroid sheet or Nicol, the intensity of the emerging beam remains uniform, then the incident light is circularly polarized. A similar result would be obtained if the incident light is ordinary unpolarized light. The two cases may be distinguished by inserting a quarter wave plate in the path of light before it falls on the analyzer. If the original light is circularly polarized, it may be considered as resultant of two coherent plane-polarized waves, that is e-ray and o-ray, which are out of phase by 90°. If the light passes through the quarter wave plate, an additional phase difference of 90° is introduced between the e-ray and o-ray. Therefore, the total phase difference becomes 180° between the e-ray and o-ray. On emerging from the quarter plate, the e- and o-rays combine to produce plane-polarized light. Therefore, if the light coming out of quarter wave plate is examined with an analyzer, light will be extinguished twice in one full rotation of the polarizer as shown in Fig.20.50 or otherwise the incident light is unpolarized.



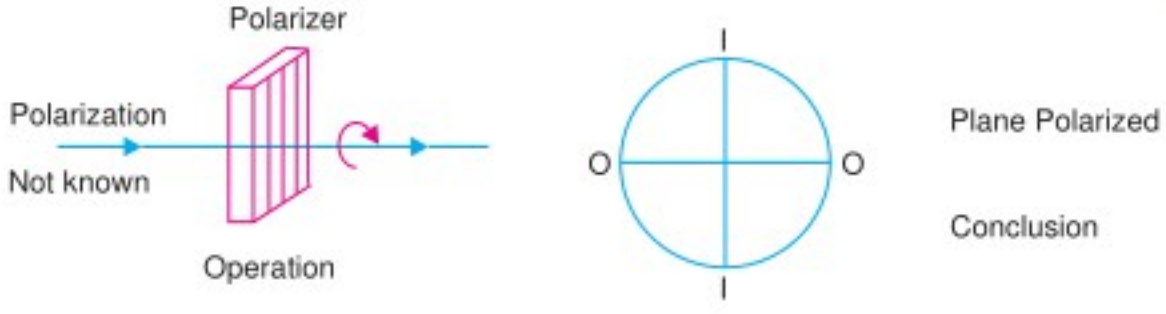
Detection of circularly polarized light

Fig. 20.50

20.22. ANALYSIS OF POLARIZED LIGHT

In practice light may exhibit any one of the three types of polarization, or may be unpolarized or a mixed type. The unaided eye cannot distinguish the different types of polarization. However, using a polarizer and a quarter wave plate, the actual type of polarization of a light beam can be ascertained. The following steps are used in the analysis of the type of polarization.

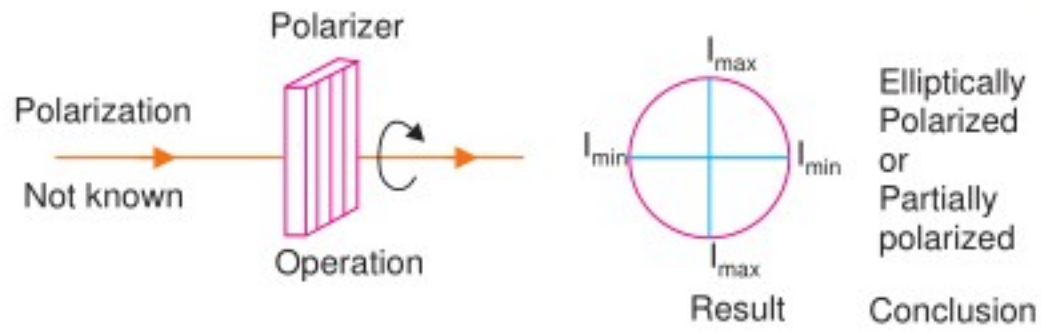
The light of unknown polarization is allowed to fall normally on a polarizer. The polarizer is slowly rotated through a full circle and the intensity of the transmitted light is observed. If the intensity of the transmitted light is extinguished twice in one full rotation of the polarizer, then the incident light is *plane polarized.*



(*a*)

Fig. 20.51

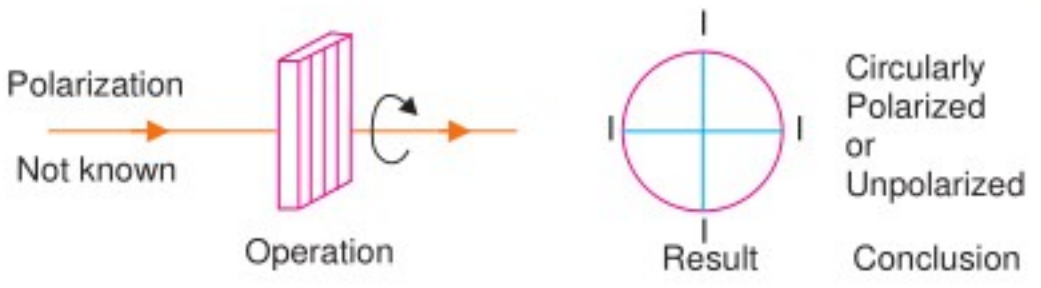
(*i*) If the intensity of the transmitted light varies between a maximum and a minimum value but does not become extinguished in any position of the polarizer, then the incident light is either elliptically polarized or partially polarized.



(*b*)

Fig. 20.51

(*ii*) If the intensity of the transmitted light remains constant on rotation of the polarizer, then the incident light is either circularly polarized or unpolarized.

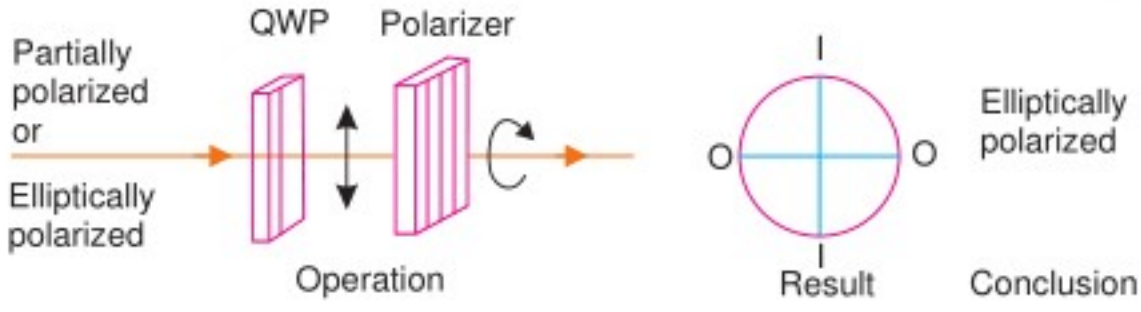


(*c*)

Fig. 20.51

To distinguish between elliptically polarized and partially polarized or between the circularly polarized and unpolarized light, we take the help of quarter wave plate. The light is first made to be incident on the quarter wave plate and then it passes through the polarizer.

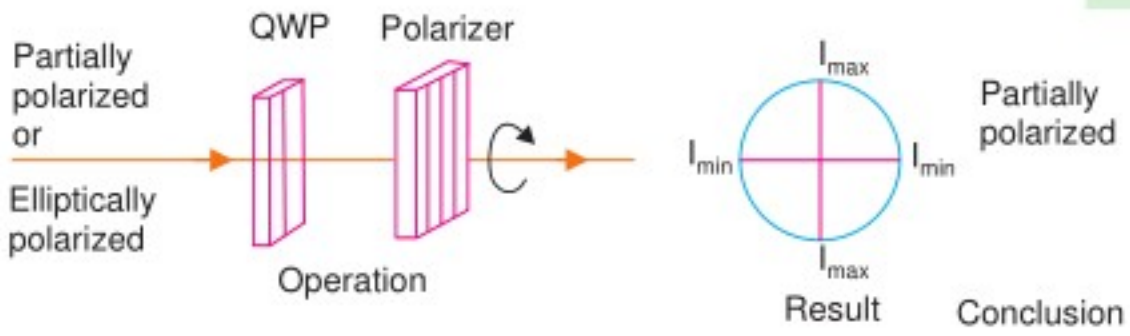
(*iii*) If the incident light is elliptically polarized, the quarter wave plate converts it into a plane polarized beam. When this polarized light passes through the polarizer, it would be extinguished twice in one full rotation of the polarizer.



(*d*)

Fig. 20.51

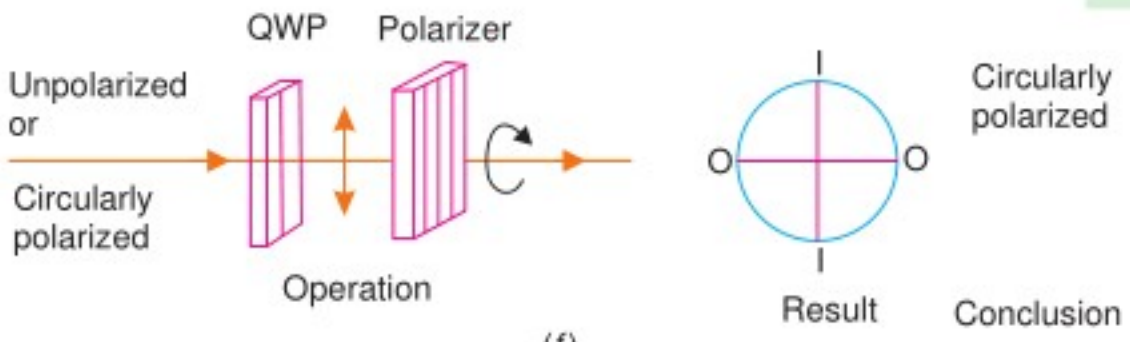
On the other hand, if the transmitted light intensity varies between a maximum and a minimum without becoming zero, then the incident light is partially polarized.



(*e*)

Fig. 20.51

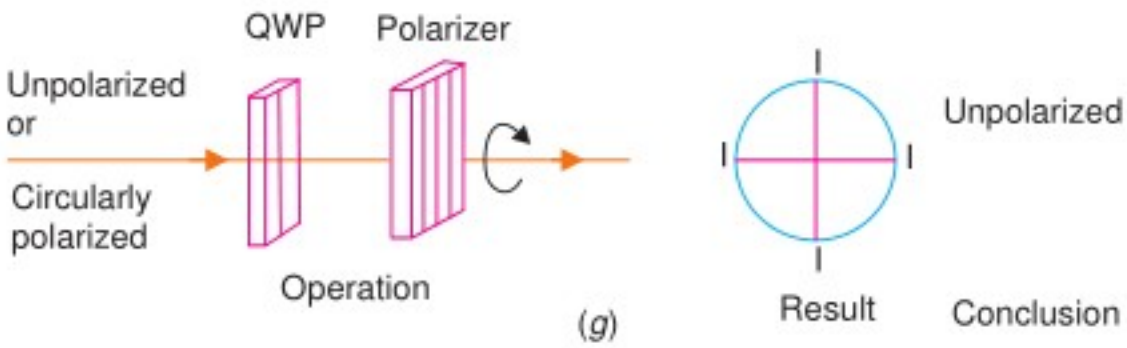
(*iv*) If the incident light is circularly polarized, the quarter wave plate converts it into plane polarized light. When this plane polarized light passes through the polarizer, it would be completely extinguished twice in one full rotation of the polarizer.



(*f*)

Fig. 20.51

On the other hand, if the intensity of the transmitted light stays constant, then the incident light is unpolarized.



(*g*)

Fig. 20.51

20.23. BABINET COMPENSATOR

A *compensator* is an optical device whose function is to compensate a path difference. It is used in conjunction with a polarizer and analyzer combination to investigate elliptically polarized light. The compensator helps in determining the axis of the ellipse and the ratio of their lengths. Elliptically polarized light may be considered as the resultant of two coherent plane polarized vibrations occurring in mutually orthogonal directions and with an initial path difference of $\lambda/4$. When the elliptically polarized light passes through the device such that it introduces a further path difference of $\lambda/4$. The total path difference between the perpendicular vibrations becomes $\lambda/2$ and the vibrations recombine after emerging from the device to produce plane polarized light. From the analysis of this plane polarized light, the information regarding the incident elliptically polarized light can be obtained.

20.23.1 CONSTRUCTION

The Babinet compensator is made of two wedge-shaped quartz sections, ABC and ADC, having equal acute angles. The wedges are placed against each other such that they form a small rectangular block as shown in Fig.20.52. One of the quartz wedges is fixed and the other can be displaced along their plane of contact with the help of a micrometer screw arrangement. Thus, the combination acts as a plate of variable thickness.

The optic axis of the first section is parallel to its refracting edge AB and the optic axis of the second section is in a direction perpendicular to the edge. The two optic axes are perpendicular to each other and also to the perpendicular to the incident beam.

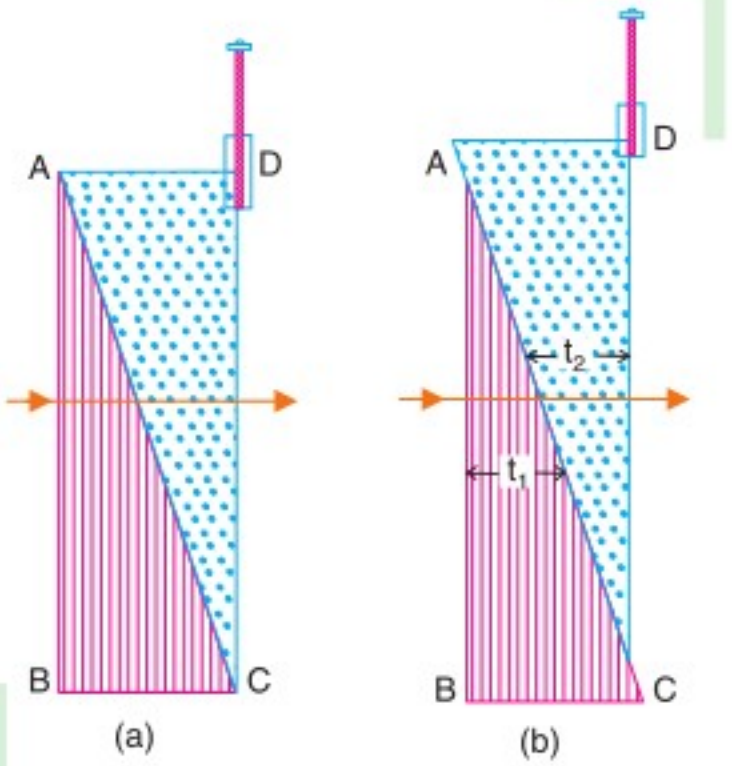


Fig. 20.52

20.23.2 PRODUCTION OF POLARIZED LIGHT

Let plane polarized light be incident normally on the face AB of the compensator. It splits into e-ray and o-ray parallel and perpendicular to AB respectively. The e-ray travels slower than o-ray in the first section, since quartz is a positive uniaxial crystal. When these rays enter the second section, the e-ray becomes o-ray since the optic axis in the second section is in a direction normal to that in the first prism. Similarly, o-ray becomes e-ray. Thus, the two rays exchange their velocities in passing from one section to the other section. The net effect is that one section cancels the effect of the other.

If d_1 is the thickness of the first section, and μ_e and μ_o are the refractive indices of quartz for e- and o-rays respectively, the path difference between the e- and o- rays in the first section will be

$$\Delta_1 = [\mu_e - \mu_o]\, d_1,$$

As the principal planes of the two sections are at right angles, the e- and o- rays change their roles in going from the first section to the second section. The velocities of e-ray and o-ray are interchanges and if the thickness of the second section is d_2, then the path difference between the rays in the second prism will be

$$\Delta_2 = [\mu_o - \mu_e]\, d_2.$$

As the compensator is thin, the separation of the rays is negligible. The net path difference between the two rays after emerging from the crystal will be

$$\Delta = \Delta_1 + \Delta_2$$

$$\Delta = (\mu_e - \mu_0)d_1 + (\mu_0 - \mu_e)d_2$$
$$= (\mu_e - \mu_0)(d_1 - d_2) \quad \ldots (20.52)$$

The net phase difference is

$$\delta = \frac{2\pi}{\lambda}(\mu_e - \mu_0)(d_1 - d_2) \quad \ldots (20.53)$$

For a ray passing through the centre of the compensator where $d_1 = d_2$ the net path difference and hence the phase difference is zero. It means that the effect of one wedge is exactly cancelled by the other. This is true for all wavelengths and the incident vibration is transmitted as such. Plane polarized light incident on the compensator will emerge as plane polarized light with its plane of vibration parallel to that of the incident light.

Any desired thickness difference $(d_2 - d_1)$ can be achieved at the centre of the compensator by moving the second section relative to the first section. Thus, any desired value of phase difference can be obtained between the e and o- rays. Therefore, the light emerging will be either plane or circular or elliptically polarized light, depending on the phase difference.

Thus the compensator has the same effect as that of a wave plate of varying thickness. The advantage of compensator is that it can be arranged to suit any wavelength while a quarter wave plate is designed to suit only one particular wavelength.

20.23.3 ANALYSIS OF ELLIPTICALLY POLARIZED LIGHT:

Using the compensator, one can determine the characteristics of elliptically polarized light.

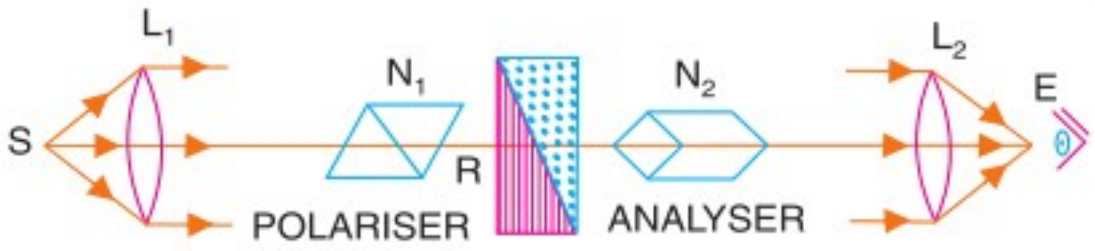


Fig. 20.53

Let the compensator be placed between crossed polarizer P and analyser A, as shown in Fig.20.53. Let the transmission axis of polarizer be oriented at 45° with respect to the optic axis of wedge ABC of the compensator. At midpoint R the light emergent from the compensator is plane polarized in the same plane as transmitted by P and therefore it will be extinguished by the analyser A. Similarly, at distances from the midpoint for which the retardation is $1\lambda, 2\lambda, 3\lambda, \ldots\ldots m\lambda$, and the emergent light is plane polarized in the same plane as transmitted by P and hence will be extinguished by the analyser. So the field of view is crossed by a series of equidistant parallel dark bands.

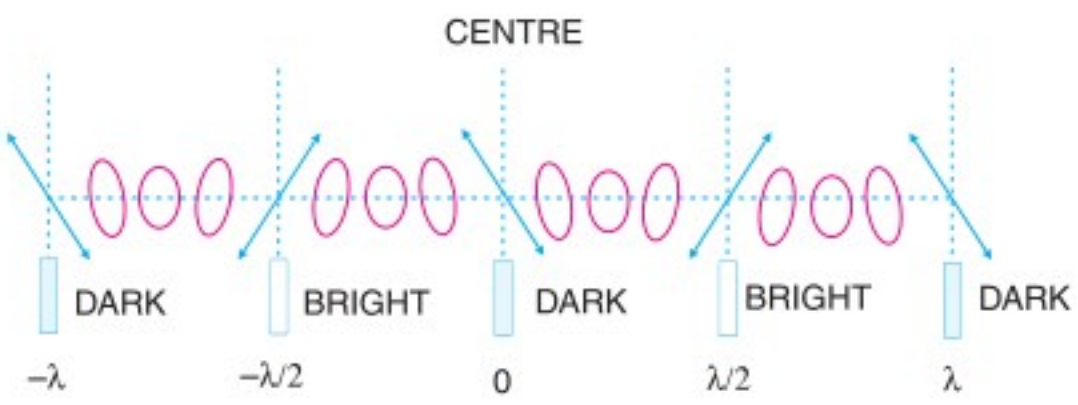


Fig. 20.54

At positions between them, where the path difference corresponds to an odd multiple of $\lambda/2$, i.e.,$\lambda/2$, $3\lambda/2$, $5\lambda/2$, ….$(2m+1)\lambda/2$, the transmitted light is plane polarized. The analyser transmits the light completely and those regions will be bright. In all other cases, the emerging light is elliptically polarized with varying parameters of the ellipse, as shown in Fig.20.54. If white light is used, the central band will be dark while others will be coloured.

By using white light source, the compensator is adjusted such that the central dark band is under cross wire and the micrometer reading is noted. The micrometer screw is turned through an angle such that the compensator introduces a phase difference of $\pi/2$ at cross wire. Then elliptically polarized light is made to be incident on the compensator. The central dark band undergoes a shift with respect to the cross wire. The compensator is rotated through an angle α in its own plane until the central

dark band is on the cross wire. The axes of the incident elliptically polarized light are parallel to the optic axes of the wedges of the compensator.

Phase difference:

The elliptical vibration can be regarded as made of two mutually perpendicular linear vibrations, which are having a phase difference, δ. δ can be determined as follows.

First the compensator is illuminated with white plane-polarised light and the micrometer is adjusted to bring the central dark band on the cross-wires. The white light is then replaced by elliptically polarised light. The central band shifts to a point where the original phase difference δ between the two component vibrations of elliptical polarised light is exactly balanced by the phase difference introduced by the compensator. This phase difference is determined by rotating the screw until the central dark band is again on the cross-wires. If this rotation is ϕ, then

$$\frac{\delta}{2\pi} = \frac{\phi}{\alpha} \text{ or } \delta = \frac{2\pi\phi}{\alpha} \quad \ldots (20.54)$$

Position of axes:

The position of the major and minor axes of the given elliptical vibration can be found as follows. The compensator is illuminated with white polarised light and the micrometer screw is adjusted to bring the central dark band on the cross-wires. The screw is then turned through an angle $\alpha/4$ so that the compensator introduces a phase difference of 90°. The central dark band now not on the cross wires. The elliptically polarised light is made incident on the compensator. Then the compensator is rotated in its own plane until the central dark band again comes on to the cross-wires. The axes of the incident light are parallel to the optic axes of the wedges of the compensator.

Ratio of the axes:

Referring to the Fig.20.55, OA and OB represent the optic axes of the two wedgews of the compensator. OC is the direction of the principal section of the analyser. DE is the direction of vibration of light emerging from the compensator at the cross-wires. The tangent of the angle θ that the principal section of the analyser makes with the optic axes of the wedge gives the ratio of the axes. Thus,

$$\tan\theta = a/b$$

θ is determined by rotating the analyser until the bands disappear and the field becomes uniformly illuminated. The angle of rotation is θ.

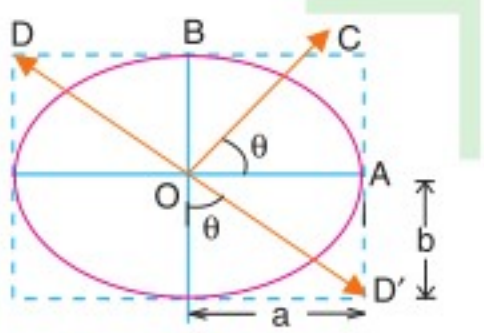


Fig. 20.55

Advantages: A quarter wave plate produces a fixed phase difference between o-ray and e-ray and can be used only for monochromatic light of one particular wavelength. In case of a compensator, the phase difference between the rays can be varied continuously and hence a compensator, made of a combination of wedges, can be used for light of any given wavelength.

20.24. FRESNEL'S RHOMB

Fresnel designed a rhomb of glass whose angles are 54° and 126° as shown in Fig.20.56.Its functioning is based on the fact that a phase difference of $\pi/4$ is introduced between the component vibrations when light is totally internally reflected back at glass - air- interface when the angle of incidence is 54°.



Fresnel's Rhomb.